

Carry Trade Dynamics under Capital Controls:

The Case of China

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Abstract:

Interest rate differentials between China and the rest of the world provide an attractive target for currency carry trade strategies, but remains problematic due to existing capital controls. We focus on copper holdings as an asset used to facilitate the carry trade. Using a unique dataset of copper stock holdings in Shanghai Futures Exchange, we study whether stock are held for carry trade or consumption purposes and how the copper carry trade position, proxied by copper stock value, reacts to the risk-return characteristics. Using an autoregressive distributed lag model, we reach three main conclusions. First, copper trade financing and stock are related to carry trade return, facilitating the Chinese carry trade. Second, copper carry trade positions are related to factors that affect return, including the onshore-offshore interest rate differential and the USD/CNY forward premium. For every 1 basis point increase in the onshore-offshore interest rate differential, copper carry trade positions increase by \$1.5 million USD. Third, traders appear unconcerned about risk factors. FX volatility between RMB/USD makes no contribution to the modeling of copper carry trade position, meaning the carry traders are either fully hedged on FX risks, or they are unconcerned about FX risks. The findings imply that potentially lower Chinese interest rates may significantly reduce Chinese demand for copper and traders are profiting from the currency hedge in the form of fixed exchange rates.

Key Words: carry trade, capital controls, copper, commodities

JEL Codes: F31, F32, F34, F37, F38, G15

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Introduction

The high interest rate in the China onshore market makes the Renminbi a good target for the currency carry trade strategy, but remains problematic due to capital controls. While the State Administration for Foreign Exchange (SAFE) and the People's Bank of China (PBOC) now allow legal channels for physical investment (FDI), and small quota-based portfolio investment via QFII, those flows are difficult to use as a source of carry trade capital. However, the gravity of large interest rate differentials pulling money into the carry trade with 'Chinese characteristics' is undeniable. Underneath the formal channels within the controlled capital account, many mechanisms exist for executing the carry trade in China.

Under a an asymmetric currency control regime, where foreign exchange current account trades are relatively free but trade in capital is strictly regulated, firms profit from substituting trade factors. In our case, firms seek to execute currency trades via the less regulated current account market in the quasi-financial asset of copper stock, but subsequently utilize the revenue as capital account holdings. We track those activities via copper stock holdings in Shanghai and their relationship to onshore-offshore interest rate differentials. Given the evidence of unregulated capital flows, we seek to answer two questions. First, due to the implied capital flows in copper trade financing, are these flows for carry trade, capital flight or other purposes? Second, if copper stock is for carry trade purposes, how do traders react to the risk-return profile of the carry trade strategy.

Using VECM and ARDL models, copper stock in China is found to be driven by carry trade activities. For every one basis point increase in the onshore-offshore interest rate differential, the bonded copper stock value increases by \$1.5 million USD. Due to the importance of the Shanghai copper holdings for the global copper market, any unwinding or change in interest rate differentials will have significant impact on global commodity market pricing and trading. We find causality running from carry trade returns to copper holdings, but that copper is only used as a medium to facilitate the carry trade and does not impact carry trade returns. At the same time, China carry traders are either perfectly hedged or unconcerned about risks.

This paper is divided into four section. We begin by describing the carry trade with Chinese characteristics. We then turn to outlining our methodological strategy given

unique data concerns. Next we build a theoretical model capturing carry trade dynamics and profitability with both revenue and risk factors. Finally, we present our results including numerous data analysis and various robustness tests. Our research presents strong evidence that Chinese copper stocks are being used primarily to facilitate a carry trade under capital controls.

The Carry Trade with “Chinese Characteristics”

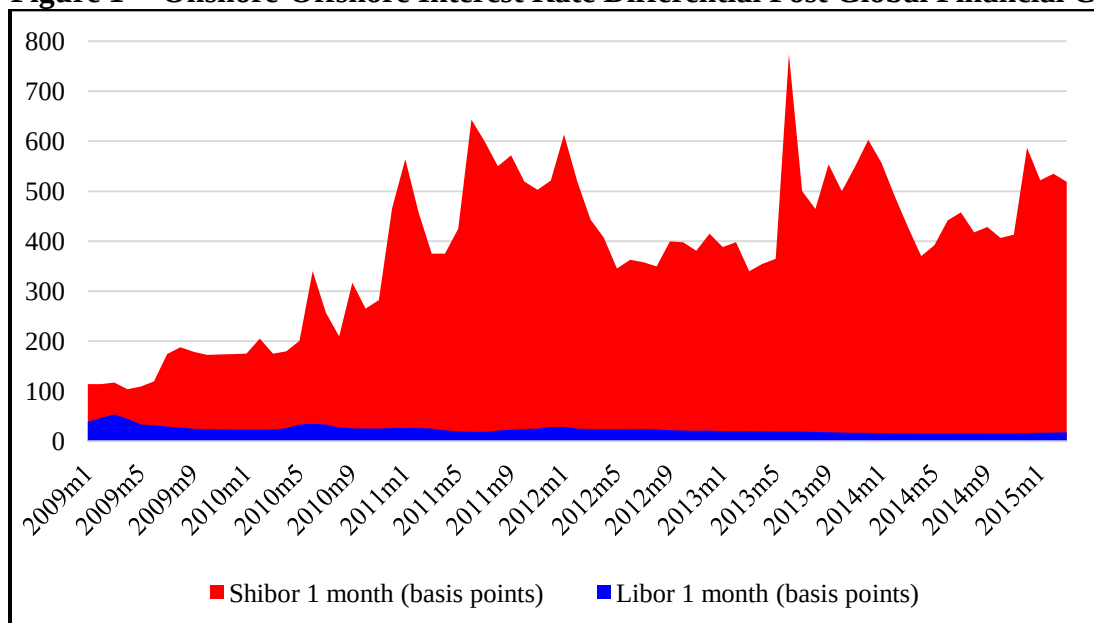
The carry trade is a common trade between countries where an investor borrows in low interest rate currencies and invests in high interest rate currencies. According to “uncovered interest parity” (UIP) theory, this strategy is supposed to be un-profitable. UIP predicts that the high interest rate currency should depreciate enough to remove gains from this strategy, under the assumption of risk neutrality and rational expectations. However, empirical studies consistently show the opposite of UIP. Research has documented evidence that high interest rate currencies have the *opposite* tendency to appreciate and low interest rate currencies tend to depreciate, which is also known as the “forward premium puzzle” (Hansen and Hodrick 1980 and Fama 1984).¹ In recent years, repeated research has empirically tested the violation of UIP finding persistent excess carry trade returns across developed and developing countries (Chinn 2005, Chaboud 2005, Darvas 2009, Flood 2002, and Bansal 2000). Recent studies on carry trade returns attempt to explain it as foreign exchange (FX) risk compensation (Burnside 2011, Lustig 2011, Menkhoff 2012, and Galati 2007). Though typically viewed as having no real impact on the real economy, some research has found that it can have an indirect pass through impact through financing channels on the broader macroeconomy (Hattori and Shin 2009). There has as of yet been no satisfactory explanation for the long term persistence and profitability of carry trade returns in financial markets.

A primary assumption of the carry trade is that it takes place in free international

¹ “Forward Premium Puzzle” is a different way to describe the same phenomenon as the “Excess Carry Trade Return”. “Forward premium puzzle” refers to the fact that forward premium tends to be greater than the spot rate appreciation from now to forward maturity. This also means that forward rate is not an unbiased estimator of future spot rate. One thing to note is that the covered interest rate parity is tested to hold at daily and lower frequencies by Akram, Rime and Sarno (2008), so the forward premium is approximately two countries’ interest rate differences with the same duration. Thus, deviations of future spot rate from forward rate means profitable carry trade strategy.

capital markets, where investors can trade financial assets in a near instantaneous manner. Recent research finds that the carry trade can be executed within common currency areas with differing risk profiles of sovereigns (Acharya and Steffen 2015). The carry trade in absence of capital controls has garnered significant research interest but there is no known study on the carry trade with ‘Chinese characteristics’. Interestingly, research indicates that currencies that are in “less financially open economies” provide good hedging against global risk (Habib and Stracca 2012). The Renminbi seems an attractive currency for the carry trade, due to significant interest rate differentials between onshore and offshore markets. For instance, the one month SHIBOR has exhibited a persistent differential with the one month LIBOR as shown in Figure 1. Over the entire post-crisis period, the onshore offshore interest rate difference, represented by the shadows in the figure, stays consistently positive, averaging 358 basis points from January 2009 to March 2015.

Figure 1 -- Onshore-Offshore Interest Rate Differential Post Global Financial Crisis

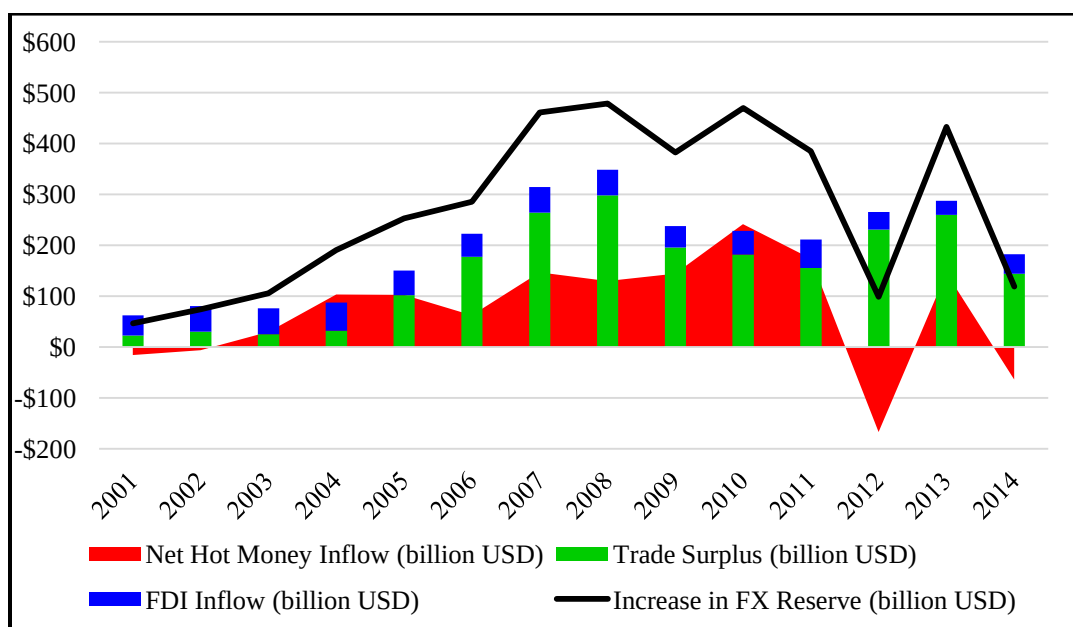


Data Source: Bloomberg and Wind

China's capital account remains controlled, but there are reasons to believe less regulated cross-border capital flows exist. China's capital controls, like similar regimes, struggle to stop or restrict the flow of capital looking for arbitrage opportunities. One study of Chinese controls on capital flows conclude that the policy has been unable to stop the capital inflows (Ma and McCauley 2004). It is generally believed that China's

long-standing presence balance of payment (BoP) errors and omissions (E&O) account is evidence of capital flows. One way to estimate the flow of capital is the formula: “foreign exchange reserves – FDI - the trade surplus”. Using this measurement, the capital flow is exhibited in figure 2 below. Capital flows peaked in 2010, with a net inflow of \$ 241.4 billion. That same year witnessed a \$181.5 billion trade surplus and FDI inflow of \$ 46.7 billion. At the same time, the E&O is not insignificant with the 2014 BoP report suggesting an E&O balance of \$140.1 billion.

Figure 2 – Net China Hot Money Inflow



Date Source: Wind and SAFE report.

There is evidence that capital flows are moving via unregulated channels facilitated by mechanisms such as copper trade financing and import/export over-invoicing. Those traceable activities offer useful information for the study of the Chinese carry trade and capital flows.

There is significant evidence for shadow capital flows via unofficial channels. Less regulated capital flows are fulfilled primarily via three channels: import and export over-invoicing, commodity financing, and the black market. Two specific channels are possible to track: international trade over-invoicing and commodity trade financing. The first channel, international trade over-invoicing, has a history of working as a channel of capital movement ever since SAFE relaxed renminbi convertibility controls. In order to control renminbi convertibility under the BoP accounts, SAFE requires proof of the

purpose for each payment and only allows current account foreign exchange transactions for goods and services payments. For example, SAFE requires banks to collect clients' supporting documents to verify goods trade. Typical supporting documents for a normal trade consist of invoice, customs declaration form, or sales contract. For instance, companies engaged in international trade have the ability to move capital from offshore to onshore via export over-invoicing.² This practice is an especially convenient way to move capital with two entities under the same multi-national group that are trading with each other, similar to the tax avoidance strategy of transfer pricing. A second channel for current account cross-border capital flow is commodity trade financing. Commodity financing is where a firm borrows at offshore interest rate, with a commodity import letter of credit from banks, with no restrictions on fund usage.³ Commodity trade financing is believed to have increased rapidly as a facilitator of the carry trade after 2008, benefiting from the widened onshore-offshore interest rate differentials.

Commodity trade financing is the tool most commonly used for carry trade activities. First, commodity trade financing is a more flexible tool compared to other flows in the capital account, in the sense that it has no restrictions on capital use and proceeds due to policy arbitrage between the relatively free current account and the heavily regulated capital account. Second, compared to over-invoicing, commodity trade financing entities are companies engaged in commodity trading, and over-invoicing entities are companies engaged in trade of other products. Companies involved in commodity trading usually have a professional trading desk and sometimes a research team. They have buy-side expertise with financial products and markets, and know how to take advantage of risk-free interest rate differentials between markets. Meanwhile, trade companies have more experience in dealing with product trades covering manufactured goods. Third, copper, and other commodities which facilitate the carry trade, while considered trade in goods and services under the current account, have the dual ability to act as a quasi-capital asset with a clear regularly updated market price that allows it to act as

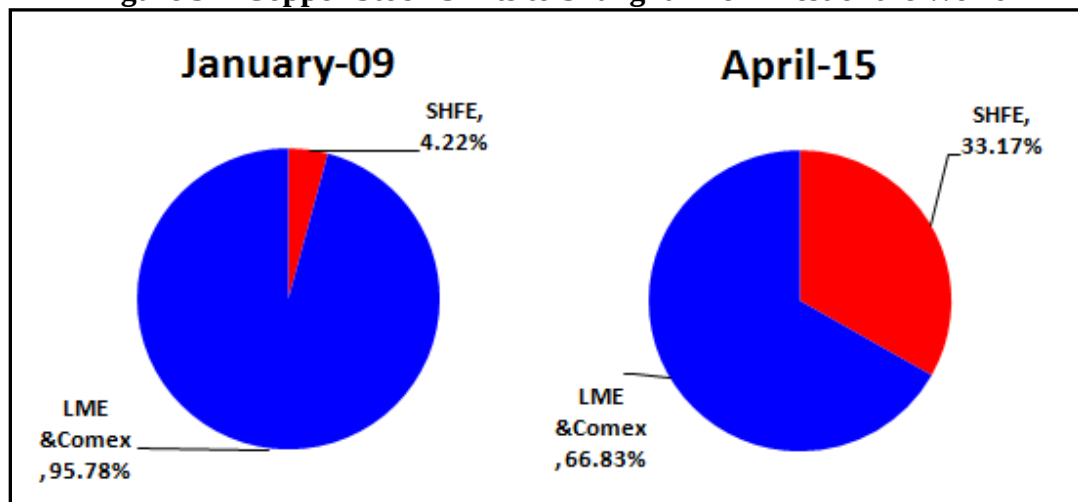
²Of course, the transaction can work in both directions depending on where the parties want to transfer the money to either jurisdictions with or without capital controls. Over invoicing allows the parties to transfer money outside of China while underinvoicing would allow the parties to transfer money *into* China.

³It should be noted there are a large number of variations and legal structures on the general concept outlined here.

collateral better than other assets.

Among all types of commodities used in the carry trade, we focus on copper for several reasons. First, copper is the most favored metal underlying commodity trade financing due to copper's durability and function as a store of value. Second, the restrictions on copper import and export are less, compared to precious metals, such as gold and silver. Third, copper is believed, though there is no statistical or survey data proving this, to be the most used commodity trade financing asset in practice. Since China's need for copper as a tool for carry trade emerged after 2008, copper stock has shifted to Shanghai from the rest of the world.

Figure 3 -- Copper Stock Shifts to Shanghai from Rest of the World



Data Source: Wind, Bloomberg, SHFE, LME and COMEX

Figure 3 shows Shanghai inventory increased from 4% of global stock in 2009 to 38% in 2014.⁴ Over the same period, the industrial production did not show evidence of a drastic change that requires such a large supporting change in inventory. Using Johansen Cointegration Test suggests no long-run relationship between SHFE copper stock value and downstream industrial production.⁵ Therefore, it is likely that carry trade is driving a significant portion of this shift in copper storage. While some research finds a negative relationship between interest rates and commodity inventories, that relies on open capital

⁴ Although a report of Credit Suisse suggests the copper stock share of Shanghai for the same periods are 30% and 75%, we could not replicate their data from Wind and Bloomberg. We make no assertions about their findings, only that using the publicly available data from Wind and Bloomberg, we could not replicate their figures.

⁵ See table 2 in Appendix 3 for results between copper stock and copper consumption.

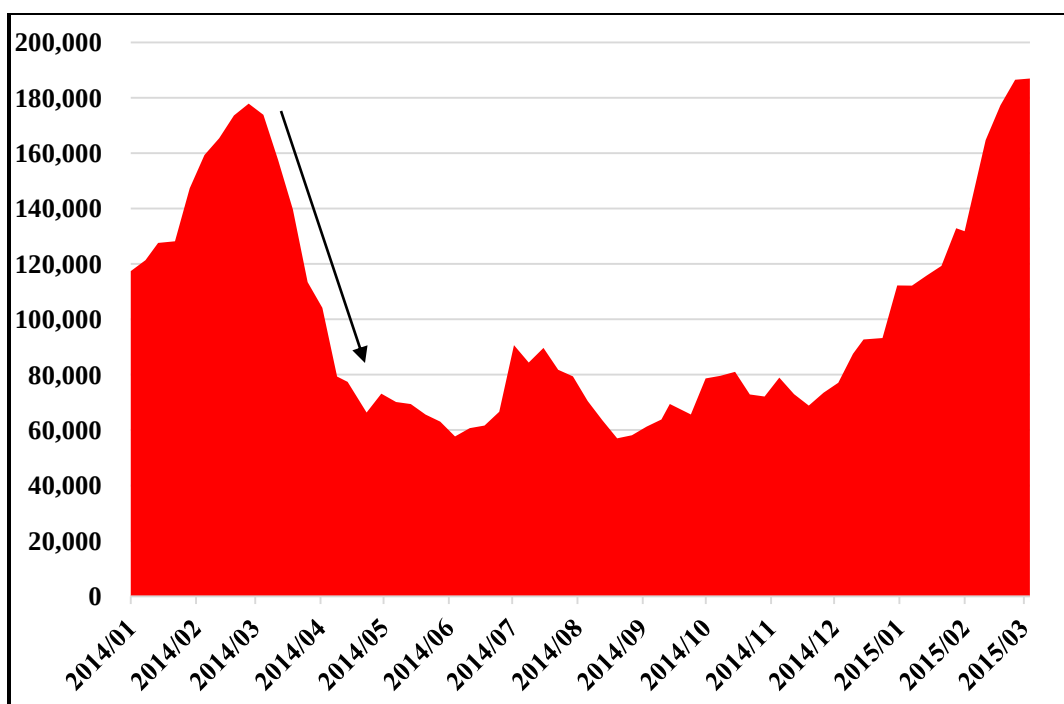
markets where in our case, the commodity is facilitating the act of the carry trade under capital controls (Frankel 2014). In section 2, we build a dynamic model to investigate whether Chinese copper stocks are driven by carry trade return. If it was a tool for carry trade, then we will model the long run and short-run relationship between copper carry trade positions and the risk-return profile.

Copper trade financing gained attention when SAFE investigated a Qingdao bonded warehouse in May 2014. The investigation was not on copper trade financing itself, but rather warehouse companies illegally issuing stock warrants amounting to several times the real stock value, increasing the financed amount or implied leverage.⁶ Based on the graph below, copper financing dropped rapidly before recovering recently. The copper stock in SHFE decreased from \$1.4 billion U.S. at its peak March 2014, to \$580 million U.S. at lowest point in May 2014 within a two month period. Later, after the investigation, copper stock returned to pre-investigation levels of \$1.4 billion U.S. in Mar 2015.⁷

Figure 4 -- SHFE Total Stock during Qingdao Investigation (in metric tonnes)

⁶ Information collected from news report, including: “Still no copper stocks shock-wave from Qingdao scandal: Andy Home” Reuters, “SAFE: Qingdao case is not only a foreign exchange fraud.” People.cn, “A rethink on Qingdao” Lin Jianhuang, China FX.

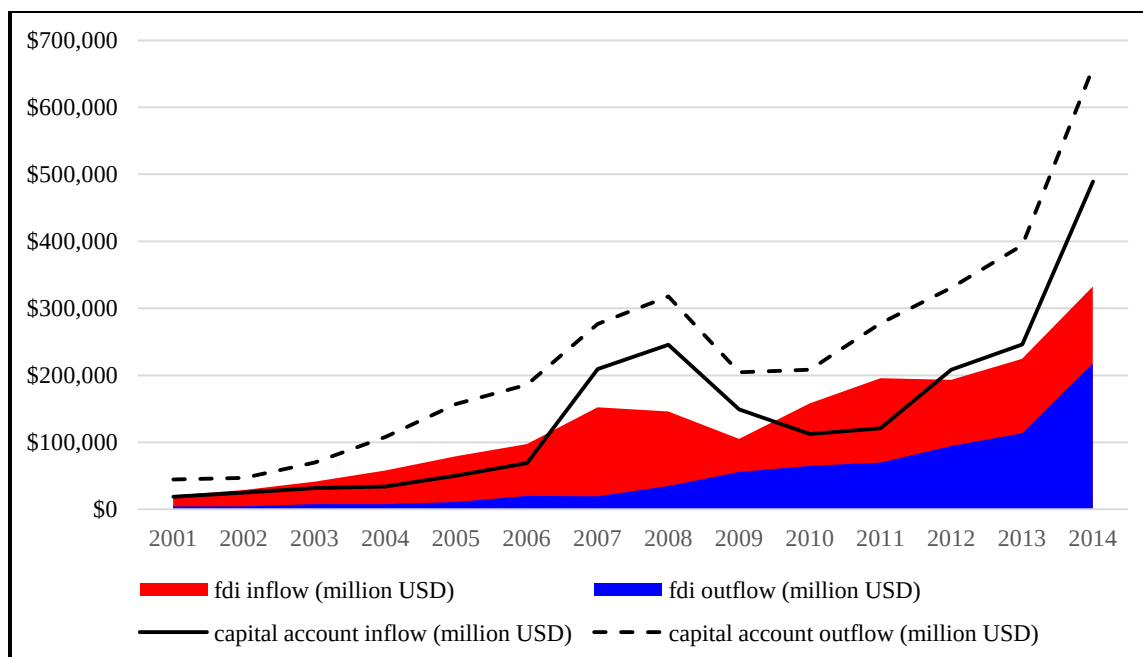
⁷ Data Source: Wind, SHFE



Source: Wind & SHFE

Restrictions on approved capital flows are significant. First, investors are only allowed to use official channels to invest in a certain types of assets and only in specified amounts. For example, FDI is restricted to project and physical investment. QDII, QFII and RQFII are restricted to certain markets, such as equity and debt. HK-SH stock connect restricts investors to specific single stocks. Second, the amount is restricted by quota. Except for FDI, which requires project registration with SAFE, QDII, QFII, RQFII and HK-SH stock connect are allotted only by government quota. The legal capital movement under capital account is closely tracked, and reported by SAFE. Here is a figure tracking its movements from 2001 to 2014:

Figure 5 -- Banks Report FX Sales/Purchases under Capital Account



Data Source: Banks Report FX Sales and Purchases reported by SAFE and available on Wind.

From the figure above, we can see that FDI is the primary component of overall capital account flow. Portfolio flows, both in and out of China, are minimal and quota driven inwards. This leaves significant amount of capital flowing into and out of China unaccounted for by traditional channels.

As an unregulated capital flow, is copper stock driven by carry trade, capital flight, or something else? Based on the evidence, we find that copper financing is driven by carry trade activity. We then seek to answer the second problem: how copper carry trade reacts to the risk-return profile of carry trade strategy.

Methodology

One challenge in empirical work with time series data is the possible spurious results due to non-stationary variables. Also, static OLS does not capture the effects that take several periods to happen. This requires a methodological approach that will properly fit the data based on the stationarity and cointegration tests, and a model that will uncover the long-run and short-run dynamic relationships at the same time.

If variables are stationary, OLS could fit the level data well, and a vector autoregressive model (VAR) will capture the feedback effect as well as effects that take periods to happen. If variables are non-stationary at level, but they are stationary after

taking first difference, or they are $I(1)$, we run cointegration tests for existence of long-run relationship. If variables are not cointegrated, it means variables are not related in the long run, and then standard OLS regression could fit the differenced data, and VAR with differenced data could capture short-run dynamics. If variables are cointegrated, then an OLS with level data will reflect the long-run coefficients, and a vector error correction model (VECM) coefficients reflect short-run dynamics adjusting to long-run equilibrium.

A special case here is a mixture of $I(1)$ and $I(0)$ variables. In this case, ARDL and Bounds Testing method could be applied (Pesaran et al.2001). This method involves several steps: First, form an unrestricted single-equation error-correction model and perform bounds testing to test for the existence of a long-run relationship. Second, if the model passes bounds testing, then we restrict the error-correction model with long-run coefficients, and estimate the short-run dynamic among variables. However, in the existence of cointegration, the most used causality test, the Granger causality test would be biased. This is because Wald tests of restrictions on the coefficients of VAR have nonstandard asymptotic properties for cointegrated $I(1)$ systems of variables (Dolado 1996). Therefore, we conducted the procedure proposed by Toda and Yamamoto to check for causality (Toda and Yamamoto 1995). The Toda-Yamamoto-Wald (TYW) test is shown to result in less Type I error probability when pretesting for cointegration (Clarke and Mirza 2006 and Zapata and Rambaldi 1997).

We proceed to empirically test whether copper trade financing is driven by the carry trade, and how the copper carry trade reacts to the carry trade risk return characteristics. We will demonstrate how copper trade financing deals are used as a carry trade facilitator, as well as the underlying risks and costs. Second, we use Johansen Cointegration Test and VECM to empirically test whether copper trade finance deals are driven by carry trade returns and test whether copper is a carry trade tool. In addition, causality relationship between copper carry trade and covered carry trade return is tested via the TY approach due to the existence of cointegration (Toda and Yamamoto 1995). Next, we use ARDL-ECM model proposed to test how the carry trade return-risk profile affects copper stocks (Pesaran 2001). Return components are onshore-offshore interest rate difference, FX forward premium (FX appreciation expectation), and the risk is FX option implied

volatility (FX rate volatility expectation). A first-tier ARDL model is built to capture the dynamics between those 3 variables and copper carry trade position. We model the carry trade position's direct reaction to FX spot rate, FX forward rate, offshore risk-free interest rate, and China onshore risk-free rate in order to better understand the the ongoing dynamics.

Theory of the Carry Trade with “Chinese Characteristics”

In practice, there are many variations of the generalized copper finance carry trade model presented here. However, in our generalized model, here are four parties involved: party A is an onshore carry trader, party B is copper owner, party C, hired or owned by A, works as a point for transit of fund, and party D is the bank that issues a letter of credit, or the offshore low interest rate credit provider. The transaction begins when onshore party A obtains a letter of credit (L/C) from onshore bank D to import copper from an offshore party B. The L/C issuance is the key in this step, which is issued at offshore USD interest rate. In order to obtain the credit, the importing party A is usually required to deposit margin capital with bank D, typically between 20% to 30% of the notional amount of the L/C. The “copper” that party A imports is frequently only a warrant on bonded copper or inbound copper.⁸

Next, copper seller B presents proper copper trade documentation such as a bill of lading, to bank D's offshore entity, and bank D pays copper seller B full amount payment of copper trade value P as per the letter of credit as credit provided for copper importer A. After the first payments have been made, the onshore party A re-exports the copper to the offshore party C and receives USD or offshore Renminbi. The transaction is fulfilled by sending the warrant documentation, without moving physical copper in bonded warehouse “offshore”.⁹ At the same time, offshore party C pays onshore copper seller A in USD or offshore Renminbi. Using proper documentation proving the copper sale under current account regulations is a normal trade, the onshore bank would convert the USD or offshore Renminbi into onshore CNY in compliance with SAFE regulation. At the point,

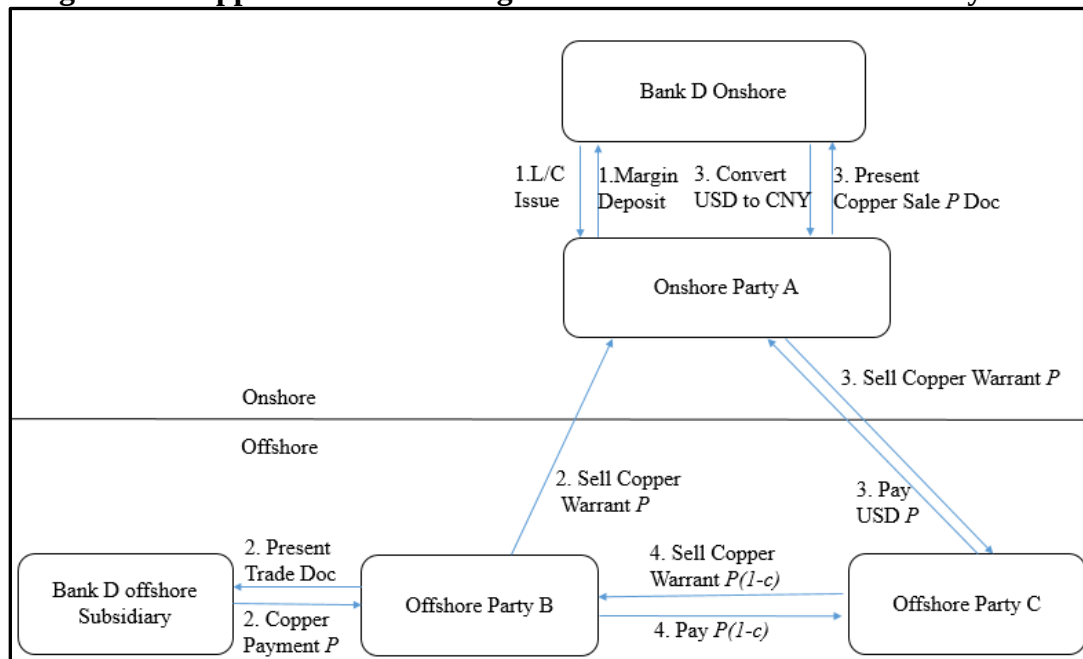
⁸ In this instance, copper is stored in warehouses in China bonded zone without entering customs, exempted from duties and fees before customs declaration. By “inbound copper” we mean copper shipped to bonded warehouse.

⁹ In many cases, A hires or owns C making it less than an “arms length transaction”.

party A has obtained funds that are not controlled by SAFE for use in offshore to onshore on credit investment, but at the lower offshore interest rate.

Finally, the offshore party C re-sells the copper back to B at a discount c from the previously mentioned trade proceed amount P that B has collected from Bank D. At this point, the copper documentation has moved through three parties without moving physical stock and is then recycled back to original owner B. The discount is in nature a fee party B charges for providing copper to assist in the transaction for party A.

Figure 6 – Copper Trade Financing Deal Structure to Facilitate Carry Trade



It should be emphasized that there are many variations of this trade. For instance, party A sometimes uses FX forward to hedge against USD/CNY movements, known as covered carry trades. Sometimes, the trade will be rotated several times during the L/C period or leverage may be used. Some deals may involve the actual physical movement of copper though value-added taxes and duties will be charged at customs declaration in this case. In some deals, party A immediately sells the copper in the China spot market. In summary, the profitability of copper trade financing deals are, in a general model, subject to the following factors.

- 1) Onshore risk-free interest rate;
- 2) Offshore risk-free interest rate;
- 3) FX spot rate:

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- 4) Risk for uncovered trades;
 - 5) FX forward premium for covered trades;

6) all other fees that are a fixed fraction of the money inflow amount: L/C fee and margin costs required by banks, fund transfer bank transaction costs, discount value paid to party B for recycling copper and value added taxes and duties if the underlying copper needs to enter customs.

To proxy for carry trade return, we assume a copper collateralization time horizon warrant with one cycle considering the case of covered and uncovered carry trades. Finally, we discuss the 1-month carry trade return as a proxy for overall carry trade return profitability but use other longer time horizons as a robustness test.¹⁰ However, the short-term nature of carry trade and prevailing use of 1 month instruments, the one month strategy is the most frequently used in carry trade studies (Burnside 2011 and Lustig 2011).

The return of a one month covered carry trade borrowing USD at 1 month Libor, converting USD to CNY at spot and going long 1 month USD/CNY forward, investing at 1 month onshore interbank rate with a 1 month duration, and exercising a forward that converts CNY back USD at maturity is calculated below:

$$R_t^c = \frac{S_t \times (1 + i_t^{on})}{F_t} - i_t^{off} - 1$$

Where

R_t^c : Covered Carry Trade Return at time t

S_t : USD/CNY spot exchange rate at time t

F_t : USD/CNY 1 Month Forward Rate at time t

i_t^{on} : Onshore 1-month risk-free interest rate at time t

i_t^{off} : Offshore 1-month risk-free interest rate at time t

Two things should be noted about the covered carry trade return variation. First, the covered carry trade return is sensitive to USD/CNY spot rate, USD/CNY forward rate, onshore risk-free interest rate, and offshore risk-free interest rate. All factors are fixed at inception, effectively hedging the carry trade return. Traders are profiting from carrying policy hedged-financially unhedged position due to the quasi-fixed exchange rate

¹⁰ See Appendix 4 for longer time horizons in robustness testing.

operated by the PBOC. This has been referred to in the literature as a “crude investment strategy”, though given the non-technical but quasi-fixed nature of the RMB, an unhedged currency strategy in this case may be reasonable (Jorda and Taylor 2012). Second, two factors, the onshore-offshore interest rate differential and the Renminbi appreciation/depreciation expectation in forward market, affect carry trade return at inception. Given the effective fixed rate of the CNY/USD and the low volatility of short term interest rate differentials, this provides a low risk well hedged return. Traders are profiting from what some have called the “peso problem” of low probability events which, in our case, are assumed to be hedged by the the quasi-fixed exchange rate (Burnside et al. 2011). Results we present later, indicate that traders are executing the carry trade in USD as compared to euro due to the fact that they do not need to hedge foreign exchange risk. Executing the carry trade between USD and RMB provides an implicit currency hedge.¹¹

The other carry trade strategy: uncovered carry trade with a one month duration by borrowing USD at 1 month Libor, converting USD to CNY at spot, investing at 1 month onshore interbank rate, holding 1 month, and converting CNY back to USD at spot rate at maturity is calculated as below:

$$R_t^u = \frac{S_t \times (1 + i_t^{on})}{S_{t+1}} - i_t^{off} - 1$$

Where

R_t^u : Covered Carry Trade Return at time t

S_t/S_{t+1} : USD/CNY spot exchange rate at time t/(t+1)

i_t^{on} : Onshore 1-month risk-free interest rate at time t

i_t^{off} : Offshore 1-month risk-free interest rate at time t

Three things are to be noted in above equation. First, the carry trade strategy profit and loss can be decomposed into spot FX rate, FX spot rate after 1 month, onshore risk-free interest rate, and offshore risk-free interest rate. Second, uncovered carry trade return is not determined at inception, rather, it is determined at maturity with the strategy (t+1), so the carry trader is supposed to care about both the return and the risks in the FX market.

¹¹ Given our focus on a single country, in this case China, we will not consider the benefits of diversification in the carry trade and has been studied else where (Burnside et al. 2008).

Third, the uncovered strategy profit and loss is affected by the fluctuations in FX rate during the holding period while the covered strategy is not. However, as noted previously, given the apparent execution primarily between USD and RMB, there is minimal implied volatility or evidence that this strategy is being conducted between EUR and RMB.

We obtained the data primarily from Wind but cross checked it and utilized comparison variables from third party sources like Bloomberg. We list all variables presented here:

CV: Copper stock value, calculated as “total copper stock in Shanghai×Copper price” in thousand U.S. dollars.

Ion: 1 month Onshore risk-free interest rate, proxied by 1 month interbank lending rate in basis points.

Ioff: 1 month offshore risk-free interest rate, proxied as 1 month USD LIBOR in basis points

ID: onshore-offshore interest rate difference (1 month), calculated as “Ion - Ioff” in basis points.

SPOT: USD/CNY spot exchange rate in basis points.

FWD: USD/CNY 1 month forward exchange rate in basis points.

FP: forward premium, calculated as “FWD-SPOT”, is the forward market expectation of Renminbi depreciation (note: the fact that forward premium implies Renminbi depreciation is subject to the quoting norm of using USD as base currency) in basis points.

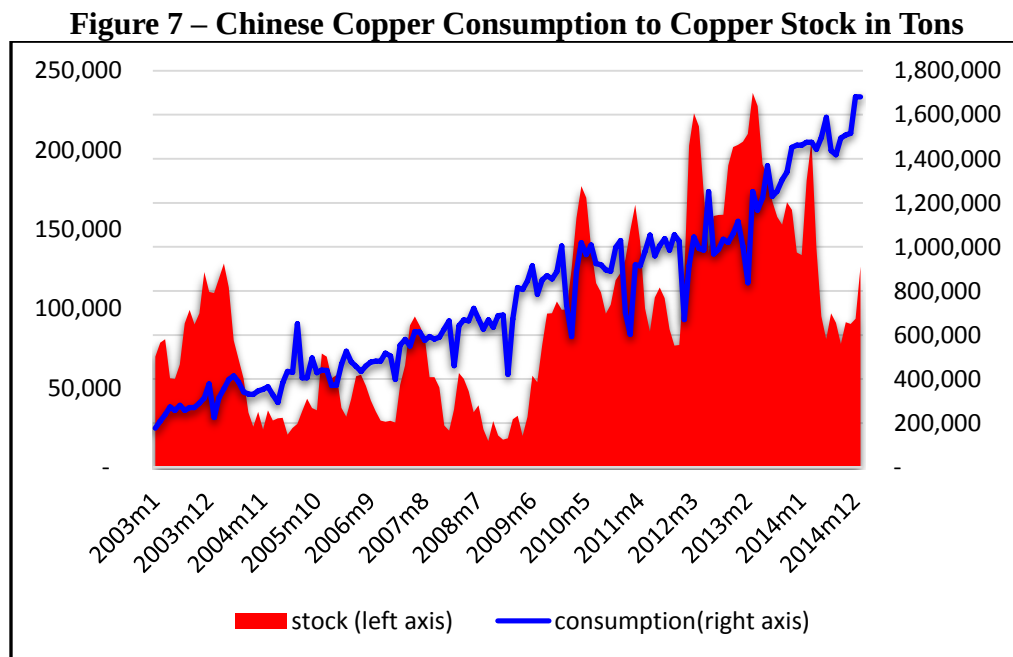
IMV: expected implied volatility over the carry trade period, proxied by 1 month FX option implied volatility in percentage points.

Rc: Covered carry trade return, calculated as “((spot USD/CNY rate) × (1+ China onshore interbank rate) / (NDF USD/CNY rate)) – Libor–1” in basis points.

We use copper stock value in Shanghai Futures Exchange as the proxy for copper carry trade positions outstanding. The reason is that commodity financing is by nature collateralized by the copper stored in a warehouse, bonded or not.¹² Another plausible

¹² There are exceptions though, when it comes to illegal multiple copper collateral deals such as those in the Qingdao bonded zone copper warehouse was investigated for issuing warrants more than once for one unit of copper

proxy is imported copper, which is used by SAFE.¹³ We believe copper stock is a better proxy than imported copper for two reasons. First, copper used for carry trade is not always imported or entering customs in a typical CCFD. In the example discussed, the copper is stored in bonded warehouses and never enters the customs process to avoid taxation. Second, imported copper includes the components consumed in industrial production. As can be seen in Figure 7, there is a clear difference between the movement of copper consumption and copper stock.



Source: Bloomberg and authors calculation

While Chinese copper consumption is clearly trend driven with noise, copper stock mimics financial asset behavior with no clear trend. We used Shanghai Futures Exchange Copper Stock Total Deliverable, reported by Shanghai Futures Exchange and collected by Bloomberg weekly from Jan 2003 to Mar 2015. The value of copper stock, CV, is calculated by multiplying by the closing price of copper price per metric ton on the copper stock reporting day. We opted to omit the storage cost of copper given its very small cost per ton per day that would not impact the final results.¹⁴

¹³ SAFE uses it in their 2014 China Cross-Border Money Tracking Report.

¹⁴ According to our research, the average storage cost per ton was 36.7 cents USD per ton per day. For a thirty day contract, this would represent approximately 0.2% of the asset value or approximately 2.47% in one year. Given then rapid turn over and implied leverage, this does not impact our final result. We thank Prof. Domenico Tariza for pointing this out.

The two primary financial factors are interest rate and exchange rate. The most appropriate onshore CNY interbank interest rate data is Shibor, the Chinese equivalent of Libor, which was first quoted on October 8, 2006, and officially introduced on January 4, 2007. In the pre-Shibor era, we used the weighted-average interbank lending rate, which follows Shibor closely in general after Shibor was introduced. One thing to note is that the forward rate we used was non-deliverable forward (NDF), which settles the price difference profit and loss in USD cash without actual delivery. There is also deliverable forward (DF) trading in the market, settled by delivery. We use the NDF price here as the DF is only allowed for current account item transactions.

There is one primary risk factor. We chose the 1 month USD/CNY FX option implied volatility to proxy for FX rate risk in the one month carry trade. The reason for choosing implied volatility over historical volatility is that FX option implied volatility is an ex-ante expectation for volatility during the carry trade period imputing regime dependence as has been noted as a significant factor in carry trade returns (Christiansen et al. 2011). Thus, it makes more sense than historical FX volatility as a measure of perceived FX risk during carry trade period others have used implied volatility (Burnside 2011, Christiansen 2011, and Menkoff 2012). Furthermore, given the low historical volatility due to the fixed exchange rate or peg with a narrow band, there is little historical volatility. Consequently, forward looking volatility is our preferred measure. Finally, we do not control for either business or market cycles during our time period. Throughout our sample, there is extremely low GDP or stock market volatility. This limits any potential insight or methodological technique that might be used to control for cyclical factors.¹⁵ We have a total of 628 data points in the sample with weekly observations from from January 1, 2003 to March 12, 2015.

Results

Now we seek to answer the question whether copper trade financing is driven by carry trade return with Johansen Cointegration Test, VECM model and Toda Yamamoto Causality Test (Toda and Yamamoto 1995). First, we want to test whether copper trade financing facilitates the carry trade. We do this by testing whether the Shanghai copper

¹⁵ The authors thank Prof. Domenico Tariza for pointing out the importance of this point.

stock is driven by carry trade return. Therefore, we raise the hypothesis:

Hypothesis 1: R_C (Covered Carry Trade Return) has long-run relationship with CV (SHFE Copper Stock Value).

We first test what kind of econometric method our data requires. To test for stationarity, we run Augmented Dicky-Fuller and Phillips–Perron unit-root tests showing that CV and R_C are both $I(1)$. The fact that our variables are cointegrated to the same order at 1 allows us to use Johansen Cointegration to discover their long run relationship. In order to find the optimal lag-length to use, we ran lag selection tests. The results of AIC, SBIC, HQIC and FPE select-order criteria suggest an optimal model lag choice of four in a VAR model.¹⁶ We use 3 lags in Johansen Cointegration Test for cointegration. Johansen-Juselius maximum likelihood method of cointegration to result shows Max-eigenvalue test and trace statistics both indicate one cointegrating vector at the 5% level with results presented in table 1 below.¹⁷

Table 1 -- Johansen Cointegration Test Results R_C (Covered Carry Trade Return) and CV(SHFE Copper Stock Value)

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximum Eigenvalue statistic	0.05Critical Value	Prob.
None	0.02	17.99	15.49	0.02	16.91	14.26	0.02
At most 1	0.00	1.078	3.84	0.30	1.07	3.84	0.30

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level

After running the Johansen Cointegration test, we conclude that there is one cointegrating vector. This indicates a long run relationship exists between copper financing and the covered carry trade return during the period January 2003 and March 2015. This result supports the first hypothesis.

We now turn to consider the magnitude of the covered carry trade return's

¹⁶ For lag selection results, please see Appendix 2.

¹⁷ The lag length here used should be $(k-1)$, since Johansen Cointegration method use differenced variable in regressions. Here, k is the optimal lag length suggested by information criteria for a VAR model.

relationship with the copper stock value. Therefore, we extract the co-integration coefficient. The cointegration estimation is displayed in table 2 below.

Table 0 Cointegrating Coefficients R_c CV

Normalized cointegrating coefficients

(standard error in parentheses)

Copper Value	Covered Carry Trade Return
1.00	(1,718.59) (406.14)

The coefficient 1,719 means that copper collateral stock goes up by \$1,719 thousand when the covered return goes up by 1 basis point. This long run positive correlation suggests that copper trade financing is driven by covered carry trade activities. Decomposing covered carry trade return, we know that Renminbi appreciation expectation in forward market and the wide onshore-offshore interest rate differentials have shifted the global copper center to Shanghai. In addition, Renminbi depreciation expectation in forward market and the narrowing onshore-offshore interest rate differentials may cause copper trade finance deals to unwind.

If we had variables integrated to the same order, a vector autoregressive or vector error correction model has several advantages. First, VAR/VECM is a system that studies two-way effects between variables in a system, modeling causality and feedback effect in the way in which one-equation models are not capable. Second, VAR/VECM tends to offer better forecasts than one-equation models. Third, VAR/VECM is useful for impulse response function analysis. Fourth, VECM manages the endogeneity problem. As CV and R_c , are both $I(1)$, we build a dynamic model to take advantage of VAR/VECM models' qualities, and study the short-run dynamic between carry trade returns and copper carry trade positions.¹⁸

Given the above discovered one vector cointegration, we built a three period lag VECM to estimate the model.¹⁹ Before any analysis of the model results, we tested serial correlation and model stability. There was no serial correlation problem shown in the Lagrange-multiplier test, and the model imposes one unit modulus, with other roots

¹⁸ Calculated with Ion, Ioff, Spot and Fwd

¹⁹ Previously selected by AIC, SBIC, HQIC, FPE and LR

strictly less than unity, suggesting model stability.²⁰

The VECM estimates are above in table 3. The error correction term coefficient is statistically significant between -1 and 0, which is ideal.²¹ The coefficient suggest the speed of adjustment to long-run equilibrium for copper value is 2%, meaning, 2% of disequilibrium from the long run equilibrium will be corrected within one period. This adjustment speed is relatively small.

Table 3 -- VECM Estimation Results

VECM Result Summary (Partial)
Dependent Variable CV

		Coef.	Std. Err.	z	P>z	[95% Conf. Interval]
	Error Correction Term	-0.02	0.01	-3.85	0.00	-0.03 -0.01
CV	lag 1	0.22	0.04	5.56	0.00	0.14 0.30
	lag 2	0.26	0.04	6.50	0.00	0.18 0.33
	lag 3	0.04	0.04	1.05	0.29	-0.04 0.12
Rc	lag1	0.98	33.45	0.03	0.98	-64.58 66.54
	lag2	34.48	33.05	1.04	0.30	-30.30 99.25
	lag3	74.65	33.01	2.26	0.02	9.95 139.35
	Constant	0.00	2206.02	0.00	1.00	-4323.71 4323.71

We then consider the carry trade return coefficient. The third lag of the covered return is statistically significant and positive at 5% level, in spite of three other insignificant lags. This means that an increase in covered carry trade return will take three weeks to make a short-term increase effect on copper trade financing. Given the required logistics for holding hard assets as the proxy for financial assets, this is reasonable result. As shown above, the copper carry trade is a complicated and time-consuming structure. For example, banks usually require weeks to process the setting up of credit line for the issuance of the letter of credit. At the same time, this result shows the CCFD has limited flexibility as a tool for carry trade.²²

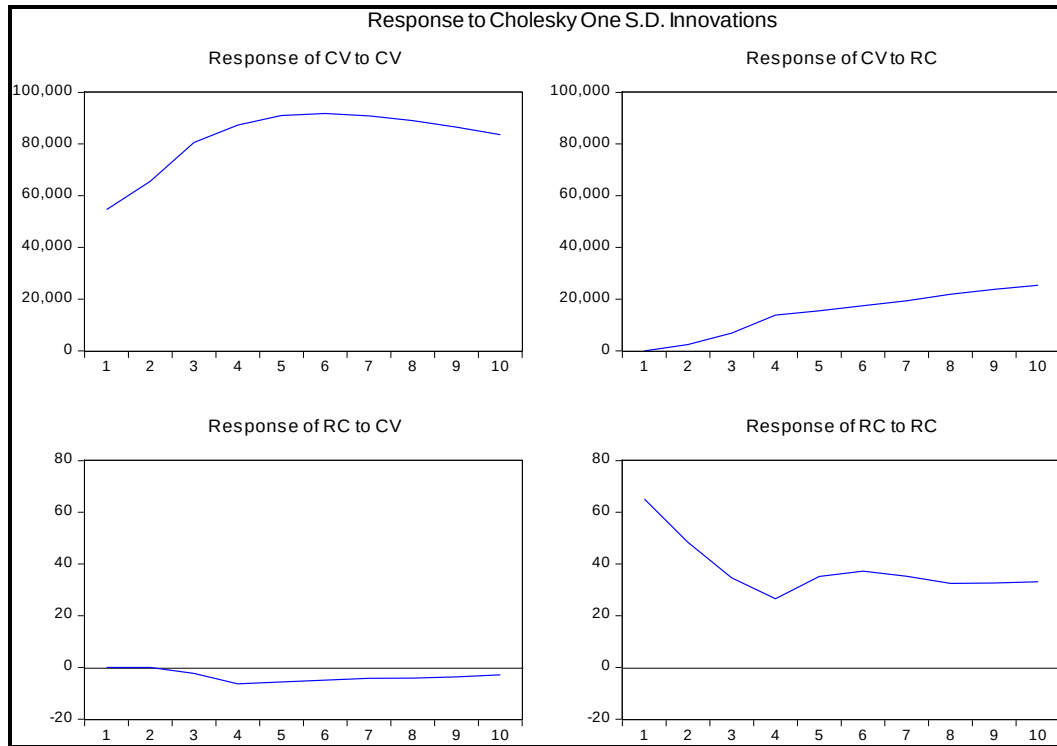
In order to investigate the dynamic relationship between covered carry trade return and copper stock value, we look at the impulse response function, to see how each variable reacts given a one-standard-error shock to a variable.

²⁰ The model has heteroskedasticity, and the normality hypothesis is rejected.

²¹ Coefficient of error correction term is the speed of adjustment to long run equilibrium. If the coefficient of error correction term falls out of the range between -1 and 0, then the model does not adjust to long run equilibrium.

²² Actually, the existence of non-zero covered carry trade return itself is a signal of an effective capital control, meaning the fund flow outside regulation scope is not sufficient to arbitrage away carry trade return.

Figure 8 -- VECM CV R^c Impulse Response Function



The impulse response function of the model with ten periods into the future is above. In the upper right graph, we can see a one-standard-error positive shock in R^c , the covered carry trade return, will lead to a slow and steady increase in CV copper stock value, over the period. In addition, in the lower left graph, we see the response of R^c to one-standard-error shock in copper stock value outstanding. We can observe a subtle negative effect that decays slowly with time. This suggests that the copper carry trade might not cause covered carry trade return changes, which are determined by interest rate and exchange rate. This leads to our second hypothesis.

Hypothesis 2: There is causality running from R^c to CV , while there is no causality running from CV to R^c .

As has been pointed out, "if two or more time-series are cointegrated, then there must be Granger causality between them - either one-way or in both directions. However, the converse is not true (Engle and Granger 1987)." Given the existence of cointegration between CV and R^c , we would expect to see at least one-way causality between these

two variables.

The problem with running the Granger causality test here is that Wald tests of restrictions on the coefficients of VAR have nonstandard asymptotic properties for cointegrated I(1) systems of variables (Dolado 1996). Therefore, we use the Toda-Yamamoto approach to non-causality to check for causality. T-Y Wald test involves the following steps. First, knowing a four period lag is optimal based on AIC, SBIC, HQIC, FPE and LR, so we run a VAR with level data utilizing 4+1 lags.²³ Second, to test for causality running from carry trade return to copper stock value, we run the Wald test with the model where copper stock value is the dependent variable. The null hypothesis is that the first four lagged coefficients are jointly zero, excluding fifth lagged coefficient.²⁴

Table 4 -- Non-causality test from R^c to CV with Toda Yamamoto Approach

Null Hypothesis: First 4 lags R^c coefficients are jointly zero

Test Statistic	Value	df	Probability
F-statistic	3.27	(4, 612)	0.01
Chi-square	13.07	4	0.01

The test result implies that we can reject the null hypothesis of Granger non-causality at the 5% level. This shows that we could use the history of covered carry trade returns to predict copper stock value. The T-Y test is consistent with the test result of one cointegration vector.

In order to test for the causality running from copper stock to covered carry trade

²³ See section preliminary results in the appendix. AIC is short for Akaike Information Criterion; SBIC is Schwarz's Bayesian information criterion; HQIC is Hannan and Quinn information criterion; FPE is Final Prediction Error Criterion; LR represents sequential modified LR test (5%). According to Toda and Yamamoto (1995), the lag length here should be calculated as $k+p$, where k is the optimal lag length of VAR model lag selection criteria, and p is the maximum order of integration of all variables. Here, we have all I(1) variables, and optimal lag is 4, so we should use 5 lags in the test.

²⁴ Here, the Wald test should include only k lags, excluding the rest p lags discussed in the previous note.

return, we again follow the T-Y approach with the other equation in the system.

Table 5 -- Non-causality test from CV to R^c with Toda Yamamoto Approach

Wald Test:

Null Hypothesis: First 4 lags CV coefficients are jointly

Test Statistic	Value	df	Probability
F-statistic	1.20	(4, 612)	0.31
Chi-square	4.80	4	0.31

The result implies that we cannot reject null hypothesis of non-causality, meaning there is no causality running from the copper stock value to covered carry trade return. Copper stock value history is not useful for forecasting covered carry trade return, and there is no feedback effect from carry trade positions outstanding to the covered carry trade return. This is logical as the returns are driven by interest rate differentials and foreign exchange risk rather than copper stocks. Additionally, the result seems reasonable, since the size of copper collateral deals is not sufficient to move currency or interest prices and arbitrage away profitability. One thing to note is that the prices we talk about here are USD/CNY exchange rate, USD interest rate, and CNY interest rate, variables unmoved by the copper carry trade. Copper financing helped drive copper stocks to \$1.42 billion USD level as of March 12, 2015. Though this could be prompted by factors such as quantitative easing or GDP performance given the weakness in the Chinese economy, a significant portion may be attributable to the copper carry trade.²⁵ This result is consistent with the impulse response function from VECM from the previous section. The TYW approach supports the hypothesis, suggesting there is causality running from carry trade return to copper carry trade position. There is, however, no causality running from copper carry trade position to carry trade return. In other words, when carry trade returns increase, copper stock in China increases but not vice versa.

As the copper stock value is driven by the carry trade, we use it as a proxy for carry trade positions to investigate the behavior of carry trade activities. To investigate this, we will build two tiers of modeling to decompose the factors that determine carry trade profit, loss and risks, in order to capture copper trade behavioral patterns. We previously

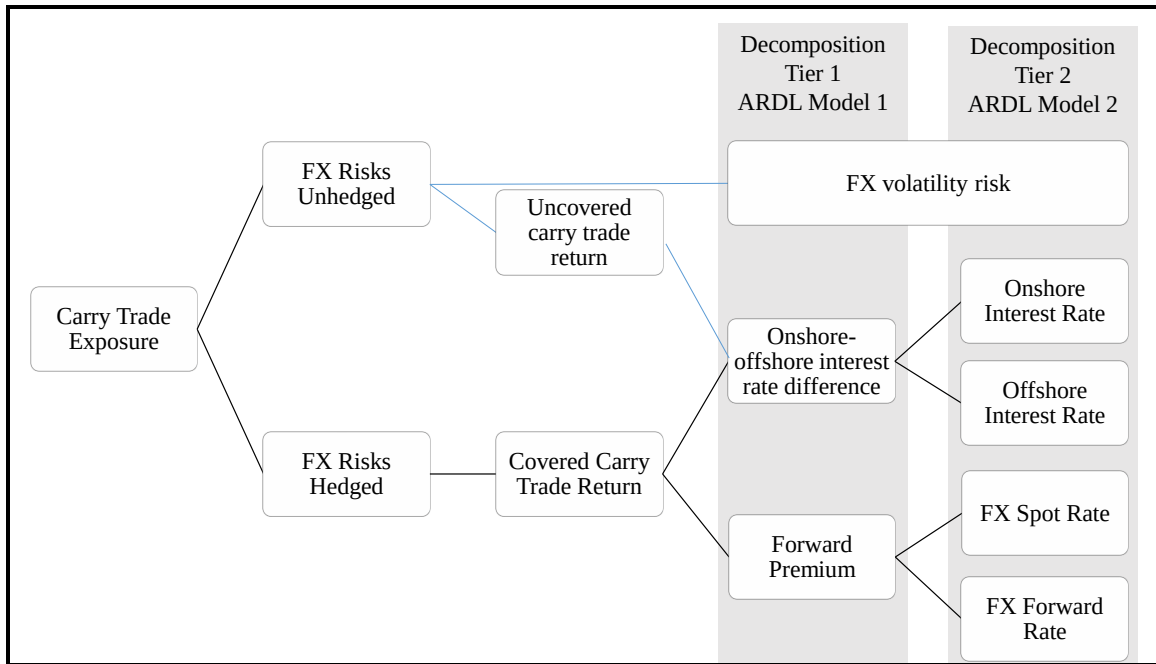
²⁵ There are many theories seeking to explain the determination of interest rate and exchange rate, which is not a focus in this paper.

mentioned that a carry trader that has fully hedged against FX rate movement has return rate: $R_t^c = \frac{S_t \times (1 + i_t^{on})}{F_t} - i_t^{off} - 1$.²⁶ The carry trader that does not hedge against FX rate movement has return rate: $R_t^u = \frac{S_t \times (1 + i_t^{on})}{S_{t+1}} - i_t^{off} - 1$.²⁷ For the entire market with a mix of hedged and unhedged carry traders, the exposure to carry trade can be decomposed into two tiers of variables to study the carry trader's reaction pattern.

²⁶ Also known as a covered carry trade.

²⁷ Also known as an uncovered carry trade.

Figure 9 -- Decomposing Carry Trade Exposure



We can see that carry traders' profit and loss are determined by three factors. Those three factors are onshore-offshore interest rate difference, FX forward premium, and FX market volatility. The first two factors decide return and the last one is risk. As the first-tier decomposition of carry trade return, we model to discover how copper carry trade positions react to those three factors. This brings us hypothesis three:

Hypothesis 3: *CV*, copper stock value, is driven by *ID*, onshore-offshore interest rate differentials, *FP* the USD/CNY forward premium, and *IMV* the USD/CNYFX option implied volatility.

We use an augmented Dicky-Fuller and Phillips–Perron unit-root test to show that *CV* and *ID* are first-order integrated or $I(1)$, while *FP* and *IMV* are stationary. Since we have a mixture of $I(1)$ and $I(0)$ variables, it is proper to use the Autoregressive Distributional Lag Model (ARDL), also known as bounds testing (Pesaran and Shin 1999 and Pesaran et al. 2001). This method is advantageous in our case for four reasons. First, ARDL allows for a mixture of $I(1)$ and $I(0)$ in the model.²⁸ Second, ARDL has only one single equation set-up, making it simple to interpret. Third, ARDL allows for different

²⁸ $I(2)$ variables cannot be modeled into ARDL, but it is our concern since we do not have any $I(2)$ variable.

lag-lengths in the model. Fourth, ARDL models can manage both the long-run cointegration and short-run dynamics.

The ARDL model is:

$$CV_t = \alpha_0 + \sum \beta_i CV_{t-i} + \sum \gamma_j FP_{t-j} + \sum \delta_k ID_{t-k} + \sum \theta_l IMV_{t-l} + \epsilon_t$$

In the formula, i, j, k, and l are the number of lags included in the model. The lag selection decision is based on information criteria. Here we use the Akaike information criteria which suggests ARDL(5,2,9,0), the Schwarz criterion suggests ARDL(3,0,0,0), and the Hannan-Quinn criterion suggests ARDL(3,2,1,0).²⁹

We have four reasons to drop IMV, implied volatility, here. First, lag selection AIC, SIC and HQ all suggest excluding ΔIMV lags. Of the top four models suggested by AIC, the top six suggested by SIC, and the top twelve models suggested by HQ all omit any lag of ΔIMV . This is a hard to neglect signal that IMV has no short run effect on CV. Second, in all of the above mentioned models, the coefficient of IMV_{t-1} term is not significantly different from 0 in t-statistics. This implies that IMV has no long run relationship with CV. Third, there is improvement in the overall significance of the unrestricted error correction model when excluding IMV. For example, the adjusted R-square of ARDL(5,2,9,0), using the optimal model suggested by AIC with IMV, is 0.988, while the adjusted R-square of ARDL(5,2,9) (i=5, j=2, k=9, excluding IMV) with optimal modeling suggested by AIC without IMV, is slightly larger though similar at 0.988. Fourth, bounds testing, presented later, fails if we include IMV, but passes when we drop IMV holding all else constant. This implies that IMV does not have a long run equilibrium or cointegration with copper carry trade position. These signs allow us to drop IMV as it suggests Chinese carry traders enter a position *without* considering IMV, which is the only risk proxy in the system of equations. There are two possible explanations. Chinese carry traders are covered carry traders, or Chinese carry traders focus exclusively on potential profit ignoring potential risks.³⁰ As previous research focusing on the yen carry trade reaction to macroeconomic surprises, we interpret our findings to mean that Chinese carry traders assign an extremely low probability to

²⁹ In this instance, i=5, j=2, k=9, and l=0, same below. The maximum number of lags considered for each variable is 12 here

³⁰ We do have to look at the result with the awareness of the fact that the trading volume of the market (Renminbi Option market) forming the volatility expectation is rather thin in the first 4 years in the sample.

macro-financial shocks as effectively hedged by the PBOC (Hutchison and Sushko 2013).

To proceed with our ARDL model, we drop the IMV variable, and estimate the equation of variables:

$$\Delta CV_t = \alpha + \sum \beta_i \Delta CV_{t-i} + \sum \gamma_j \Delta FP_{t-j} + \sum \delta_k \Delta ID_{t-k} + \mu_0 CV_{t-1} + \mu_1 FP_{t-1} + \mu_2 ID_{t-1} + \epsilon$$

Akaike information criteria suggests ARDL (5, 2, 9), the Schwarz criterion suggests ARDL (3, 0, 0), and the Hannan-Quinn criterion suggests ARDL (3, 2, 1).³¹ Note that dropping IMV does not affect the lag selection of other variables. Based on a balanced consideration of over-fitting of AIC method and coefficient significance, ARDL (3, 2, 1) is taken to further analysis. The Breusch-Godfrey Serial Correlation Lagrange Multiplier test does not reject the null hypothesis of no serial correlation of any order up to two in ARDL (3,2,1) at the 5% level. This means that there is no serial correlation in the residual. Therefore, we can undertake bounds testing with this model.

We formulate and estimate an unrestricted error-correction model is set up as follow:

$$\Delta CV_t = \alpha + \sum \beta_i \Delta CV_{t-i} + \sum \gamma_j \Delta FP_{t-j} + \sum \delta_k \Delta ID_{t-k} + \mu_0 CV_{t-1} + \mu_1 FP_{t-1} + \mu_2 ID_{t-1} + \epsilon \quad ^{32}$$

To discover the existence of long-run equilibrium we utilize a bounds test (Pesaran 2001). Bounds testing is an F-test of the hypothesis $H_0: \mu_0 = \mu_1 = \mu_2 = 0$. The lower bound is applied when all of the variables are I(0), and the upper bound is used when all of the variables are I(1). If the F-statistic is below the lower bound, it implies that the variables are I(0), so no cointegration is possible. If the F-statistic exceeds the upper bound, it means that we have cointegration. If the F-statistic falls between the bounds, the test is inconclusive. Before testing, this unrestricted error-correction model was tested by Breusch-Godfrey LM test, and cannot reject null of no serial correlation at 5% level or greater. The bounds test result is shown below in table 6:

³¹ Here ARDL(5,2,9) actually corresponds to i=4, j=1, k=8 in the above equation, because the equation is a deferred once. The maximum lag considered for each variable is 12.

³² This is almost like a traditional Error Correction Model, except for the unrestricted coefficient terms in the error-correction term. In traditional ECM, the error-correction term is $z_{t-1} = CV_{t-1} - (a_0 + a_1 FP_{t-1} + a_2 ID_{t-1} + a_3 IMV_{t-1})$, where the a's are the OLS estimates of the α 's in $CV_{t-1} = \alpha_0 + \alpha_1 FP_{t-1} + \alpha_2 ID_{t-1} + \alpha_3 IMV_{t-1} + \epsilon$. For this reason, this formula is also called "unrestricted ECM" or "conditional ECM" in Pesaran et al. (2001)

Table 6 -- ARDL1 Bounds Testing Result

ARDL Bounds Test		
Null Hypothesis: No long-run relationships exist		
Test Statistic	Value	k
F-statistic	4.98	2
Critical Value Bounds		
Significance	Lower Bound	Upper Bound
10%	3.17	4.14
5%	3.79	4.85
2.50%	4.41	5.52
1%	5.15	6.36

The F-statistic 4.97 is larger than upper bound 4.85 at 5% significance level. We reject the hypothesis of no long-run relationship, meaning there is long term relationship between copper stock value, forward premium, and interest rate differentials. The long-run relationship suggests capital control is consistent with copper trade financing as a mechanism for carry trade.

We now extract the long-run relationship between variables from the unrestricted error correction model estimated in the first step. The estimation result is shown below in table 7:

Table 7 -- ARDL1 Unrestricted Error Correction Model Results

Dependent Variable: D(CV)				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(CV(-1))	0.23	0.04	6.04	0.00
D(CV(-2))	0.27	0.04	6.91	0.00
D(FP)	-0.26	0.19	-1.38	0.17
D(FP(-1))	-0.59	0.19	-3.10	0.00
D(ID)	-31.98	32.57	-0.98	0.33
C	8946.55	3980.62	2.25	0.03
FP(-1)	-0.02	0.08	-0.23	0.82
ID(-1)	32.47	11.12	2.92	0.00
CV(-1)	-0.02	0.01	-3.67	0.00

In the above table, we extract the long-run multiplier between the dependent and independent variables. We find that copper is significant, and the long-run multiplier between copper stock value and the interest rate differential is $-(32.47/(-0.02)) = 1515.16$. This produces one of our primary findings. This follows with a variety of research

finding interest rate differentials act as the driving catalyst for the carry trade even in the face of macroeconomic shocks (Anzuini and Fornari 2012). In the long run, an increase of 1 basis point in the onshore-offshore interest differential leads to a \$1.5 million USD increase in copper stock value. This can be seen table 8 below:

Table 8 -- ARDL1 Long Run Coefficients

Dependent Variable: CV				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
FP	-0.85	3.76	-0.23	0.82
ID	1,515.16	449.80	3.37	0.00
C	417,456.81	135,188.90	3.09	0.00

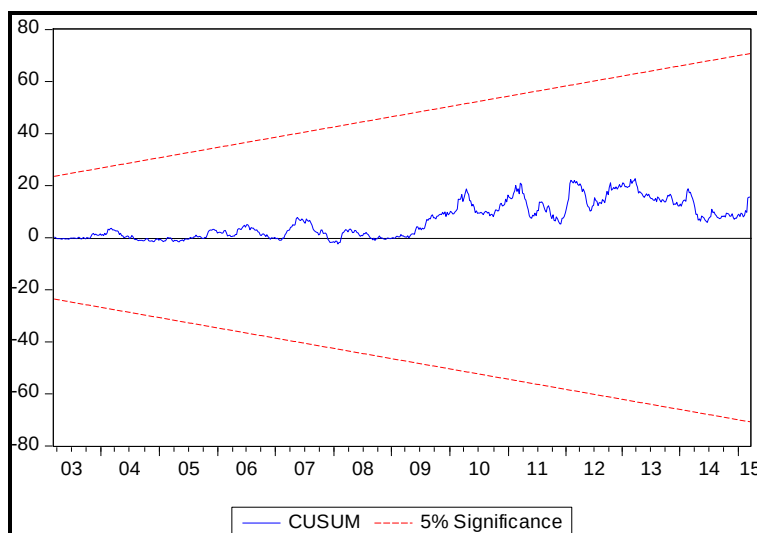
Here FP is actually insignificant, meaning that carry traders are not sensitive to the forward premium in the long run. Our long run findings are consistent with the hypothesis that onshore-offshore interest differential widening drives increases in CCFD carry trade, but is not impacted by Renminbi depreciation expectation.

Now that bounds testing results suggest the existence of co-integration, we can meaningfully estimate the restricted error correction model:

$$\Delta CV_t = \alpha + \sum \beta_i \Delta CV_{t-i} + \sum \gamma_j \Delta FP_{t-j} + \sum \delta_k \Delta ID_{t-k} + \phi Z_{t-1} + \epsilon_t$$

Where $z_{t-1} = CV_{t-1} - (a_0 + a_1 FP_{t-1} + a_2 ID_{t-1})$. Here, a_0, a_1, a_2 are the long run coefficients suggested above. The Cusum test suggests stability of the model at the 5% level or greater. The model is found stable in a Cusum test at 5% significance.

Figure 10 -- ARDL1 Restricted ECM Stability Cusum Test



The estimation result of the restricted error correction model is shown below in table 9:

Table 9 -- ARDL1 Restricted ECM Estimation Results

Dependent Variable: CV

Selected Model: ARDL(3, 2, 1)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(CV(-1))	0.23	0.04	6.04	0.00
D(CV(-2))	0.27	0.04	6.91	0.00
D(FP)	-0.26	0.19	-1.38	0.17
D(FP(-1))	-0.59	0.19	-3.10	0.00
D(ID)	-31.98	32.57	-0.98	0.33
CointEq(-1)	-0.02	0.01	-3.67	0.00

$$\text{Cointeq} = \text{CV} - (-0.8486 \cdot \text{FP} + 1515.1647 \cdot \text{ID} + 417456.8061)$$

Here, the error-correction term coefficient is significantly negative, and it is between 0 and -1, ensuring convergence in the model to a significant long run relationship. The coefficient -2.14% represents the speed of adjustment to long run equilibrium, meaning nearly 2% of any disequilibrium from the long run is corrected within one period, one week in our data. The adjustment of disequilibrium is rather slow. This finding is consistent with the adjustment speed of the main model suggested by error correction term coefficient, suggesting an adjustment speed of 2% each period week.

Now we look at the short run effect from forward premium below in table 10.

Table 10 -- ARDL1 RECM Wald Test against No Short-Run Effect from FP

Null Hypothesis: FP lags coefficients are jointly zero			
Test Statistic	Value	df	Probability
F-statistic	5.35	(2, 619)	0.005
Chi-square	10.71	2	0.005

The coefficients of ΔFP lags are both significantly negative, based on table 10 below and a Wald test against null jointly zero coefficients. Therefore, we conclude that there is short run causality running from forward premium to copper carry trade outstanding size. Moreover, the negative sign of coefficient is consistent with the hypothesis that forward premium, Renminbi depreciation expectation, causes an unwinding of copper carry trade. The short run coefficients suggests no short run effect running from interest rate differentials to copper trade financing counters our hypothesis, which is puzzling given the existence of long-run relationship. In addition, there is information lost in data with the calculation of interest rate difference (here, we calculate ID as “ $I_{on} - I_{off}$ ”). In the following part, we further de-compose carry trade risk-return profile, and build another model to capture the lost information in raw data, and to crosscheck our results.

We have tested the hypothesis 3. We find copper carry trade is driven by forward premium and interest rate differentials. Specifically, an increase of 1 basis point in the onshore-offshore interest difference leads to an estimated \$1.5 million USD increase in copper trade financing outstanding. The speed of adjustment to long-run equilibrium is rather slow. We find the only risk involved in this strategy, FX volatility, is not contributive in modeling CV given the implicit FX policy hedge. Therefore, we drop the variable, and conclude carry traders are not affected by risks.

Now we proceed to construct a second tier ARDL, to model the effect on copper carry trade from onshore interest rate, offshore interest rate, FX spot rate and FX forward rate. We do this for several reasons. First, we hope to capture the information lost in calculation of forward premium and interest rate differentials in the previous model. Second, we hope to use this model to understand the causality using more direct variables, such as LIBOR, SHIBOR and Spot USD/CNY rate to copper carry trade. This will allow us to make some forecasts about copper stock values in the future. Decomposing the

forward premium and interest rate difference, we have this following hypothesis:

Hypothesis 4: The China copper carry trade position(CV) is positively affected by onshore interest rate(*Ion*) and Renminbi forward rate(*SPOT*), but negatively affected by offshore interest(*Ioff*) rate and Renminbi spot rate(*FWD*).³³

We estimate a model to test this hypothesis as follows:

$$CV = \alpha_0 + \alpha_1 ION + \alpha_2 IOFF + \alpha_3 SPOT + \alpha_4 FWD + \epsilon$$

Augmented Dicky-Fuller and Phillips–Perron unit-root tests show *CV*, *SPOT*, *FWD* and *Ion* are I(1), while *Ioff* is I(0). Once again, we have a mixture of I(1) and I(0) variables and adopt the ARDL approach. After a balanced consideration of model significance, over-fitting and under-fitting, we adopt the model suggested by Akaike Information Criteria: ARDL(5,0,4,2,2). The Breusch-Godfrey Serial Correlation Lagrange Multiplier test does not reject the null hypothesis of no serial correlation of any order up to two in ARDL(5,0,4,2,2) at the 5% level or greater.³⁴ This means that there is no serial correlation in the residual. Therefore, we undertake bounds testing with this model.

We estimate the unrestricted error correction model, and run the bounds testing with the results shown below in table 11.

Table 11 -- ARDL2 Bounds Testing

ARDL Bounds Test		
Null Hypothesis: No long-run relationships exist		
Test Statistic	Value	k
F-statistic	3.59	4
Critical Value Bounds		
Significance	Lower Bound	Upper Bound
10%	1.9	3.01
5%	2.26	3.48
2.50%	2.62	3.9
1%	3.07	4.44

The F-statistic 3.49 is larger than upper bound 3.48 at 5% significance level or greater. Consequently, we reject the hypothesis of no long-run relationship. We conclude there is long run relationship in this model and proceed to extract the long-run relationship from

³³ Here we drop IMV by default because it was dropped in the previous model at set-up

³⁴ For test results, please go to Appendix 2 for serial correlation results.

the unrestricted error correction model below in table 12.

Table 12 -- ARDL2 Long Run Cointegration Coefficients

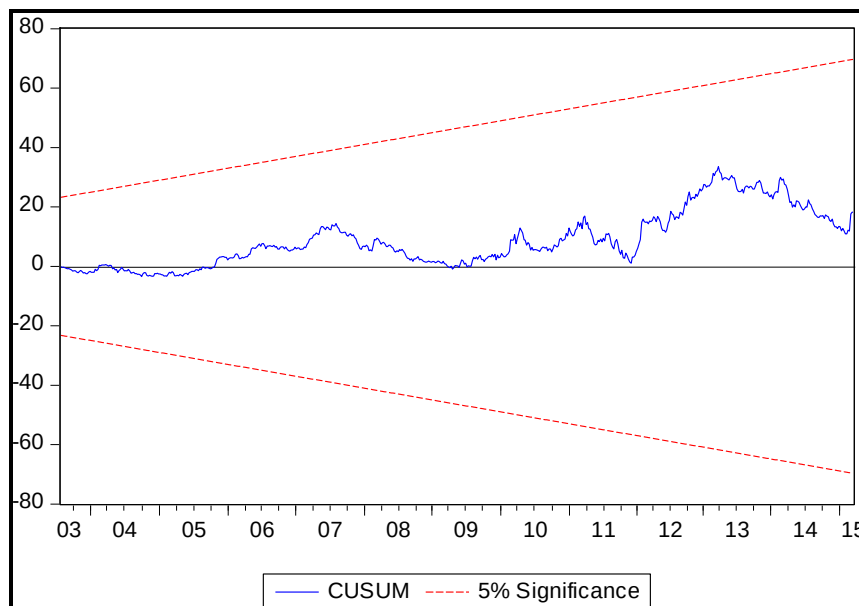
Long Run Coefficients Dependent Variable: CV

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LIBOR	(1,187.57)	610.44	-1.95	0.05
SHIBOR	2,130.52	625.61	3.41	0.00
SPOT	-31.20	334.94	-0.09	0.93
FWD	32.64	335.20	0.10	0.92

For every 1 basis point increase in onshore interest rate, the copper carry trade position will increase USD 2,130,520. Meanwhile, for every 1 basis point increase in Libor, copper carry trade will unwind for value of USD 1187,570. Both coefficients are consistent with our hypothesis. However, spot and forward of USD/CNY is not significant in the long run, which goes against the hypothesis that FX rates affects copper carry trade in the long run. However, given the implied currency hedge, this is not an entirely surprising result.

Now that bounds testing results suggest the existence of co-integration, we can meaningfully estimate the regular restricted error correction model to discover the short run dynamics. The Cusum test suggests stability of the model at 5% level or greater:

Figure 11- ARDL2 Restricted Error Correction Model Cusum Test



The restricted ECM estimate result is shown below in table 13:

Table 13 -- ARDL2 Restricted Error Correction Model Estimation Results

Dependent Variable: CV

Selected Model: ARDL(5, 0, 4, 2, 2)

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(CV(-1))	0.21	0.04	5.41	0.00
D(CV(-2))	0.25	0.04	6.15	0.00
D(CV(-3))	0.03	0.04	0.77	0.44
D(CV(-4))	0.09	0.04	2.24	0.03
D(Ioff)	-28.64	15.15	-1.89	0.06
D(Ion)	7.57	35.75	0.21	0.83
D(Ion(-1))	-65.42	42.32	-1.55	0.12
D(Ion(-2))	-31.25	42.56	-0.73	0.46
D(Ion(-3))	81.79	36.01	2.27	0.02
D(SPOT)	2.25	23.04	0.10	0.92
D(SPOT(-1))	53.01	22.36	2.37	0.02
D(FWD)	-29.04	19.45	-1.49	0.14
D(FWD(-1))	-61.25	19.51	-3.14	0.00
CointEq(-1)	-0.02	0.01	-4.06	0.00

$$\text{Cointeq} = \text{CV} - (-1187.5667 * \text{LIBOR} + 2130.5174 * \text{SHIBOR} - 31.2044 * \text{SPOT} + 32.6437 * \text{FWD})$$

Here, the error-correction term coefficient is significantly negative, and it is between 0 and -1, ensuring convergence to a significant long run relationship. The coefficient -2% represents the speed of adjustment to long run equilibrium, meaning nearly 2% of any disequilibrium from the long-run is corrected within one period, one week in our dataset. This adjustment speed is slightly faster than the previous model but still relatively slow, after we take in more factors into the system. We tested for autocorrelation with Breusch–Godfrey method, and found no autocorrelation.³⁵

Looking at the short run coefficients, we can conclude the effects of Ion, Ioff, SPOT, and FWD separately. First, the Wald Test against the null that all short-run coefficients of Ion are jointly zero does not reject the null at 5% and above, meaning there is no effect running from onshore interest rate to copper carry trade in short-run adjustments.

³⁵ Please see Appendix 2 table 4 for results.

Table 14 -- ARDL2 RECM Wald Test against No Short-Run Effect from from Ion

Wald Test:

Null Hypothesis: Ion lags coefficients are jointly zero

Test Statistic	Value	df	Probability
F-statistic	1.85	(4, 612)	0.12
Chi-square	7.41	4	0.12

Looking at coefficients of lags separately, we have a mix of positive and negative signs in coefficients of Ion. Only the third lag is significant, and the coefficient is the largest in scale, at 81.79, meaning a basis point increase in onshore interest rate at this period will cause around 82 thousand U.S. dollar increase in copper carry trade position. Given the total size of copper stock holdings, this is an economically small result. This is interesting to see that it takes 3 weeks for onshore interest rates to have the most significant effect on copper carry trade deals. Given the length of time required to enter into the copper carry trade, this is a reasonable finding. Second, Ioff is significantly different from zero at 10% or greater, meaning there is short-run causality running from offshore interest rate to copper carry trade positions. Third, Wald Test against the null that all short-run coefficients of SPOT are jointly zero rejects the null at 5% and above, meaning there is a short-run effect running from Renminbi spot rate to copper carry trade.

Table 15 -- ARDL2 RECM Wald Test against No Short-Run Effect from SPOT

Wald Test:

Null Hypothesis: SPOT lags coefficients are jointly zero

Test Statistic	Value	df	Probability
F-statistic	3.17	(2, 609)	0.04
Chi-square	6.34	2	0.04

The first lag of the spot rate is significant and positive at the 5% level, meaning that it takes one week for spot rate to influence copper carry trade positions. Fourth, Wald Test against the null that all short-run coefficients of FWD are jointly zero rejects the null at 5% and above, meaning there is short-run causality running from forward to copper carry trade.

Table 16 -- ARDL2 RECM Wald Test against No Short-Run Effect from FWD

Wald Test:

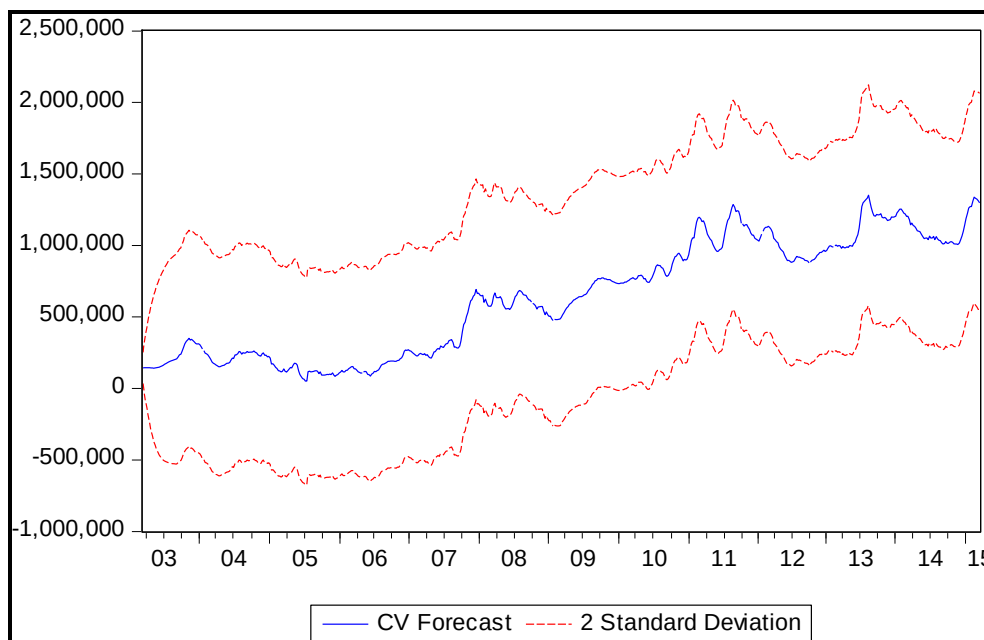
Null Hypothesis: FWD lags coefficients are jointly zero

Test Statistic	Value	df	Probability
F-statistic	5.87	(2, 609)	0.00
Chi-square	11.74	2	0.00

The first lag of the spot rate is significant and negative at 5% level. The coefficient of -61.25 in table 13 suggests that for every basis point increase in the Renminbi forward rate or Renminbi depreciation expectation, copper carry traders will decrease their carry trade position by an estimated \$61 thousand USD. Here it is interesting to note that both spot and forward will take one week to impact copper carry trade positions.

A forecast could be made on the copper carry trade position with our model. The forecast for future copper stock value will be fluctuating and slightly uptrending given expected wide interest rate differentials.

Figure 12 -- ARDL2 Copper Stock Value Forecast



We conclude that in the long run, the copper stock is driven by onshore and offshore interest rate in the equilibrium. Onshore interest rate drives up copper carry trade position, and offshore interest rate drives down copper carry trade position. FX rates, both spot and forward are not significant, which is consistent with FP being not significant in the long

run from previous model. We find FX rate is not driving copper stock value in the long run. However, the FX rates have effects in short run, and the direction is consistent with our hypothesis.

Conclusion

In conclusion, we find copper trade finance is a tool for carry trade, and import & export over-invoicing a tool for capital flight. Moreover, copper carry trade position is driven by carry trade return, but no risks involved. 1 basis point increase in onshore-offshore interest rate differential, copper carry trade position increases by 1.5 million USD in the long run. Domestic monetary acceleration causes capital flight, and capital outflows have negative effect on monetary expansion.

The thesis seeks to answer three questions: are unregulated capital flows in copper trade finance deal and over-invoicing, for carry trade, capital flight or other purposes? Second, if the capital flow is for carry trade, how would it react to the risk-return profile of the strategy? Third, how those capital flows affect domestic monetary supply?

Given a mixture of I(1) and I(0) time series, we use VECM and ARDL-ECM model in attempts to answer those three questions. First, Johansen cointegration test suggest copper trade finance is long-run related to carry trade return, therefore a carry trade tool. Specifically, the TYW causality test result suggests causality running from carry trade return to copper position, but copper carry trade is not capable of removing positive carry trade return, determined by a series of interest rates and exchange rates. Pesaran (2001) ARDL bounds testing suggest there is no long run relationship between covered carry trade return and over-invoicing net capital outflow, meaning over-invoicing is not a tool for carry trade. Another piece of evidence to support this is, over-invoicing consistently suggest net capital outflow in our observation despite high RMB carry trade profitability, meaning capital outflow is likely to be a capital flight tool.

For the second problem, ARDL bounds testing shows that copper carry trade positions, in the long run, is related to return factors including onshore-offshore interest rate difference, the USD/CNY forward premium, onshore interest rate, offshore interest rate, USD/CNY spot rate and forward rate (Pesaran 2001). For every 1 basis point increase in onshore-offshore interest rate differential, copper carry trade position

increases by 1.5 million USD in the long run.. VECM short-term adjustment coefficients suggest that it takes 3 week for covered carry trade return to take effect on copper carry trade position, reflecting the inflexibility of copper trade finance deals for carry trade. However, risks do not affect carry traders. Risk proxy FX volatility gives no significant contribution to the ARDL modeling of copper carry trade position, meaning the carry traders are either fully hedged on FX risks, or do not care about FX risks.

For the third problem, ARDL model results suggests an increase in China M2 has long run relationship with over-invoicing capital flows, copper carry trade position and legal capital flows. One interesting finding here is ARDL long-run cointegration coefficients suggest M2 incremental increases with capital flight via over-invoicing, while short-run coefficient suggest the opposite. One explanation is that long-run coefficient is a mixture of two-way causality, and short-run coefficient is just a one-way causality from over-invoicing to M2. Therefore, the case could be that acceleration in M2 causes capital flight, while capital flight causes M2 incremental to decrease. This explanation is supported in a further VECM impulse response function experiment and a Toda Yamamoto causality test.

References

- Acharya, V. & Steffen, S. (2015). The “greatest” carry trade ever? Understanding eurozone bank risks. *Journal of Financial Economics*, 115(2), 215-236
- Akram, Q. F., Rime, D., & Sarno, L. (2008). Arbitrage in the foreign exchange market: Turning on the microscope. *Journal of International Economics*, 76(2), 237-253.
- Anzuini, A. & Fornari, F. (2012). Macroeconomic Determinants of Carry Trade Activity. *Review of International Economics*, 20(3): 468-488
- Bansal, R., & Dahlquist, M. (2000). The forward premium puzzle: different tales from developed and emerging economies. *Journal of International Economics*, 51(1), 115-144.
- Bouvatier, V. (2010). Hot money inflows and monetary stability in China: how the People's Bank of China took up the challenge. *Applied Economics*, 42(12), 1533-1548.
- Burnside, C., Eichenbaum, M., Kleshchelski, I., & Rebelo, S. (2011). Do peso problems explain the returns to the carry trade?. *Review of Financial Studies*, 24(3): 853-891.
- Burnside, C., Eichenbaum, M., Rebelo, S. (2008). Carry Trade: The Gains of Diversification. *Journal of the European Economic Association*. 6(2-3): 581-588
- Chaboud, A. P., & Wright, J. H. (2005). Uncovered interest parity: it works, but not for long. *Journal of International Economics*, 66(2), 349-362.
- Chinn, M. D., & Meredith, G. (2005). *Testing uncovered interest parity at short and long horizons during the post-Bretton Woods era* (No. w11077). National Bureau of Economic Research.
- Chipalkatti, N., & Rishi, M. (2001). External debt and capital flight in the Indian economy. *Oxford Development Studies*, 29(1): 31-44.
- Christiansen, C., Rinaldo, A., & Söderlind, P. (2011). The time-varying systematic risk of carry trade strategies. *Journal of Financial and Quantitative Analysis*, 46(04): 1107-1125.
- Claessens, S., Naude, D., & Mundial, B. (1993). *Recent estimates of capital flight*. Debt and International Finance, International Economics Department, World Bank.
- Clarke, J. A., & Mirza, S. (2006). A comparison of some common methods for detecting Granger noncausality. *Journal of Statistical Computation and Simulation*, 76(3): 207-231.
- Darvas, Z. (2009). Leveraged Carry Trade Portfolios. *Journal of Banking and Finance*, 33(5): 944-957
- Dolado, J. J., & Lütkepohl, H. (1996). Making Wald tests work for cointegrated VAR systems. *Econometric Reviews*, 15(4): 369-386.
- Engle, R. F., & Yoo, B. S. (1987). Forecasting and testing in co-integrated systems. *Journal of Econometrics*, 35(1): 143-159.
- Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319-338.
- Fang, Z. Y. H. P. P., & Ming, X. (2007). An Empirical Analysis on Short-term International Capital Inflow and its Motivation in China: Based on Triple-arbitration Model [J]. *Studies of International Finance*, 9, 008.
- Flood, R. P., & Rose, A. K. (2002). Uncovered interest parity in crisis. *IMF staff papers*, 252-266.
- Frankel, J. (2014). Effects of speculation and interest rates in a “carry trade” model of commodity prices. *Journal of International Money and Finance*, 42: 88-112
- Frankel, J. A. (2009). New estimation of China's exchange rate regime. *Pacific Economic Review*, 14(3), 346-360.

-
- Frankel, J. A., & Okongwu, C. (1995). *Liberalized portfolio capital inflows in emerging capital markets: Sterilization, expectations, and the incompleteness of interest rate convergence* (No. w5156). National bureau of economic research.
- Galati, G., Heath, A., & McGuire, P. (2007). Evidence of carry trade activity. *BIS Quarterly Review*, 3, 27-41.
- Gunter, F. R. (2004). Capital flight from China: 1984–2001. *China Economic Review*, 15(1): 63-85.
- Habib, M. & Stracca, L. (2012). Getting Beyond Carry Trade: What makes a safe haven currency? *Journal of International Economics*, 87(1): 50-64.
- Hansen, L. P., & Hodrick, R. J. (1980). Forward exchange rates as optimal predictors of future spot rates: An econometric analysis. *The Journal of Political Economy*, 88(5): 829-853.
- Hattori, M. & Shin, H. (2009). Yen Carry Trade and the Subprime Crisis. *IMF Staff Papers*, 56(2): 384-409
- Huang, H., & Shuiling, W. (2004). Exchange rate regimes: China's experience and choices. *China Economic Review*, 15(3): 336-342.
- Hutchison, M. & Sushko, V. (2013). Impact of macro-economic surprises on carry trade activity. *Journal of Banking and Finance*, 37(4): 1133-1147
- Jorda, O. & Taylor, A. (2012). The carry trade and fundamentals: Nothing to fear but FEER itself. *Journal of International Economics*, 88(1): 74-90
- Liya, L. (2008). Do Overseas "Hot-money" Push Sharpe Rise of the Chinese Stock Market and Real Estate Market?. *Journal of Financial Research*, 10, 007.
- Lustig, H., Roussanov, N., & Verdelhan, A. (2011). Common risk factors in currency markets. *Review of Financial Studies*, hhr068.
- Lütkepohl, H. (2007). *New introduction to multiple time series analysis*. Springer Science & Business Media.
- Ma, G., & McCauley, R. N. (2008). Efficacy of China's capital controls: Evidence from price and flow data. *Pacific Economic Review*, 13(1), 104-123.
- Rishi, M., & Boyce, J. K. (1990). The Hidden Balance of Payments: Capital Flight and Trade Misinvoicing in India, 1971-1986. *Economic and Political Weekly*, 1645-1648.
- Menkhoff, L., Sarno, L., Schmeling, M., & Schrimpf, A. (2012). Carry trades and global foreign exchange volatility. *The Journal of Finance*, 67(2), 681-718.
- Moreno, R. (1996). Intervention, sterilization, and monetary control in Korea and Taiwan. *Economic Review*, 23-33.
- Ouyang, A. Y., Rajan, R. S., & Willett, T. D. (2010). China as a reserve sink: The evidence from offset and sterilization coefficients. *Journal of International Money and Finance*, 29(5), 951-972.
- Pesaran, M. H., Shin, Y., & Smith, R. J. (2001). Bounds testing approaches to the analysis of level relationships. *Journal of applied econometrics*, 16(3), 289-326.
- Pesaran, M. H., & Shin, Y. (1998). An autoregressive distributed-lag modelling approach to cointegration analysis. *Econometric Society Monographs*, 31, 371-413.
- Tang, K., & Zhu, H. (2014). Commodities as Collateral. *Available at SSRN 2355674*.
- Toda, H. Y., & Yamamoto, T. (1995). Statistical inference in vector autoregressions with possibly integrated processes. *Journal of econometrics*, 66(1), 225-250.
- Xiaohua, Z. Y. S. (2008). An Empirical Research on the Appreciation of RMB, Rising of Stock Price and Hot Money Inflow [J]. *Journal of Financial Research*, 11, 010.

-
- Xuan, Y. Y. Z. (2007). Empirical Research on the Impact of Foreign Exchange Reserve on the Money Base in China [J]. *World Economy Study*,1, 009.
- Zapata, H. O., & Rambaldi, A. N. (1997). Monte Carlo evidence on cointegration and causation. *Oxford Bulletin of Economics and statistics*,59(2), 285-298.

Appendix 1 – Descriptive Statistics

	CV	FP	ID	Rc	Ru	FWD	Ioff	Ion	SPOT
Mean	616,585.80	(9,105.73)	159.47	170.79	178.79	71,396.97	168.93	328.40	71,488.03
Median	481,372.60	(9,950.00)	148.43	161.00	152.88	68,342.50	48.13	307.18	68,331.50
Maximum	1,929,687.00	87,800.00	999.03	875.66	981.26	82,785.00	582.00	1,018.33	82,775.00
Minimum	42,453.02	(143,000.00)	(324.06)	(299.71)	(326.42)	60,963.00	14.95	92.00	60,508.00
Std. Dev.	496,137.40	32,162.81	263.41	249.67	261.33	8,072.53	188.01	141.66	8,199.21
Skewness	0.78	0.25	-0.07	0.02	0.00	0.25	0.95	0.88	0.23
Kurtosis	2.51	3.62	2.32	2.43	2.37	1.45	2.35	4.00	1.44
Sum	3.87E+08	-5.72E+06	1.00E+05	1.07E+05	1.12E+05	4.48E+07	1.06E+05	2.06E+05	4.49E+07
Sum Sq. Dev.	1.54E+14	6.49E+11	4.35E+07	3.91E+07	4.28E+07	4.09E+10	2.22E+07	1.26E+07	4.22E+10
Observations	628	628	628	628	628	628	628	628	628

Appendix 2 – Lag Selection Testing

Table 1 -- Lag Selection – VECM Model Copper Stock Value and Covered Carry Trade Return

VAR Lag Order Selection Criteria

Endogenous variables: CV(Copper Stock Value) RC(Covered Carry Trade Return)

Exogenous variables: C

Included observations: 618

Lag	LR	FPE	AIC	SC	HQ
0	NA	1.02E+16	42.53	42.55	42.54
1	3927.91	1.74E+13	36.16	36.20	36.18
2	75.14	1.55E+13	36.05	36.12	36.08
3	70.71	1.40E+13	35.95	36.05	35.99
4	39.84	1.33e+13*	35.89*	36.02*	35.94*
5	5.43	1.34E+13	35.90	36.06	35.96
6	3.77	1.35E+13	35.91	36.09	35.98
7	14.11*	1.33E+13	35.90	36.11	35.98
8	3.33	1.34E+13	35.90	36.15	36.00
9	7.99	1.34E+13	35.90	36.17	36.01
10	3.56	1.35E+13	35.91	36.21	36.03

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

SC: Schwarz information criterion

HQ: Hannan-Quinn information criterion

Table 3 --VECM Residual Serial Correlation LM Test

VEC Residual Serial Correlation LM Tests

Null Hypothesis: no serial correlation at lag order h

Sample: 1/02/2003 3/12/2015

Included observations: 623

Lags	LM-Stat	Prob
1	3.69	0.45
2	3.25	0.52

Probs from chi-square with 4 df.

Table 4 -- ARDL2 BREUSCH–GODFREY Autocorrelation Test for RECM

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	0.33	Prob. F(2,604)	0.72
Obs*R-squared	0.66	Prob. Chi-Square(2)	0.72

Table 5 -- Toda Yamamoto Causality Test IMO and $\Delta M2$ Lag Selection

VAR Lag Order Selection Criteria

Endogenous variables: DM IMO

Exogenous variables: C

Included observations: 170

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-3897.9	NA	2.89E+17	4.59E+01	45.92	45.90
1	-3874.7	45.59	2.30E+17	45.65476	45.77	45.70
2	-3861.3	26.00	2.06E+17	45.54424	45.73	45.62
3	-3822.5	74.31	1.37E+17	45.1354	45.39	45.24
4	-3779.3	81.94	8.64E+16	44.67353	45.01	44.81
5	-3758.8	38.32	7.12E+16	44.4796	44.89	44.64
6	-3749.7	16.81	6.71E+16	44.4196	44.90	44.61
7	-3742.9	12.36	6.49E+16	44.38693	44.94	44.61
8	-3724.6	32.85	5.49E+16	44.21927	44.85	44.47
9	-3706.4	32.46263*	4.65E+16	44.05135	44.75229*	44.33578*
10	-3701.2	9.02	4.59e+16*	44.03789*	44.81	44.35

* indicates lag order selected by the criterion

LR: sequential modified LR test statistic (each test at 5% level)

FPE: Final prediction error

AIC: Akaike information criterion

SC: Schwarz information criterion

HQ: Hannan-Quinn information criterion

Therefore we Decided to use VAR model of 10 lags, and test first 9 lags' jointly equal zero Wald test in testing causality in 4.3.3

Appendix 3 -- Testing Johansen Cointegration between Copper Stock Value and Uncovered Carry Trade Return

In the main context, we use covered carry trade return to proxy for carry trade return as a whole, which is not precise. Here we show the result of Johansen Cointegration between uncovered carry trade return and copper stock value. Both variables are I(1), and AIC suggests 4 lags for the model, therefore, we use 3 lags for Johansen Cointegration Test.

Table 1 -- Johansen Cointegration Test Result of Ru and CV

Sample: 1/02/2003 3/12/2015

Included observations: 624

Series: CV RU

Lags interval: 1 to 3

Data Trend:	None	None	Linear	Linear	Quadratic
Test Type	No Intercept	Intercept	Intercept	Intercept	Intercept
	No Trend	No Trend	No Trend	Trend	Trend
Trace	1	1	1	1	2
Max-Eig	1	1	1	1	2

*Critical values based on MacKinnon-Haug-Michelis (1999)

The result suggest one cointegration vector based on critical value MacKinnon-Haug-Michelis (1999) at 5% or greater level for most trend/intercept assumptions. We conclude that CV is also cointegrated with RU, which is consistent with our hypothesis.

Copper Consumption and Copper Stock

We found there exists cointegration between carry trade returns and SHFE copper stock value. Still, we cannot rule out the possibility of the influence from copper industrial consumption in the absence of empirical support. We use data from China copper raw material output monthly data to proxy for domestic copper consumption. Copper raw material is upper-stream to car production, home appliance production, electric wire and other productions. The data is I(1), copper stock value, as we have shown in main text, is also I(1). We run Johansen Cointegration Test to test for long-run relationship between copper consumption and Shanghai copper stock value.

Table 2 -- Johansen Cointegration Test: Copper Consumption and Copper Stock Value

Unrestricted Cointegration Rank Test(Trace & Maximum

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximun Eigenvalue statistic	0.05Critic al Vlue	Prob.
None	0.069	10.360	15.495	0.254	10.295	14.265	0.193
At most 1	0.000	0.064	3.841	0.800	0.064	3.841	0.800

Trace test indicates 0 cointegrating eqn(s) at the 5% level

Max-eigenvalue test indicates 0 cointegrating eqn(s) at the 5% level

Max-eigenvalue and trace statistic both indicate 0 cointegrating equations at 5% level. This means there is no long-run cointegration between copper consumption and copper stock value.

Appendix 4 -- Robustness Test

In the main text, our research focused on the one month carry strategy. In other words, relying on one month credit instruments. Here we test three, six, and 12 month strategies for robustness checking. Johansen Cointegration Tests between onshore-offshore interest rate difference and copper stock value suggest long run cointegration of all three maturities. Test results are shown below:

Table 1 -- Johansen Cointegration Test: 1-year onshore-offshore interest rate difference & copper stock value

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximum Eigenvalue statistic	0.05 Critic al Vlua	Prob.
None	0.031	13.403	12.321	0.033	13.397	11.225	0.020
At most 1	0.000	0.005	4.130	0.952	0.005	4.130	0.952

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level

Table 2 -- Johansen Cointegration Test: 6-month onshore-offshore interest rate difference & copper stock value

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximum Eigenvalue statistic	0.05 Critic al Vlua	Prob.
None	0.031	13.553	12.321	0.031	13.546	11.225	0.019
At most 1	0.000	0.008	4.130	0.943	0.008	4.130	0.943

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level

Table 3 -- Johansen Cointegration Test: 3-month onshore-offshore interest rate difference & copper stock value

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximum Eigenvalue statistic	0.05 Critic al Vlua	Prob.
None	0.031	13.540	12.321	0.031	13.526	11.225	0.019
At most 1	0.000	0.015	4.130	0.921	0.015	4.130	0.921

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level

In all tests, trace statistic and maximum eigenvalue statistic both suggest the existence of 1 cointegrating equation at the 5% level for all three maturities. This is consistent with out results with 1 month strategy in main text.

Appendix 5 – EUR/CNY Carry Trade

In the main text, we focused on USD/CNY carry trade strategy. Johansen Cointegration Tests between EUR-CNY interest rate difference and copper stock value suggest long-run relationship between two variables.

Table 1 -- One month interest rate differential and copper stock value

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximun Eigenvalue statistic	0.05Critic al Vlua	Prob.
None	0.051038	23.36788	15.49471	0.0027	20.84969	14.2646	0.004
At most 1	0.006307	2.518194	3.841466	0.1125	2.518194	3.841466	0.1125

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level

Also, Johansen Cointegration Tests between EUR/CNY forward premium and copper stock value suggest long-run relationship between two variables.

Table 2 -- One month forward premium and copper stock value

Unrestricted Cointegration Rank Test(Trace & Maximum Eigenvalue)

No. of Cointegrating equation(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.	Maximun Eigenvalue statistic	0.05Critic al Vlua	Prob.
None	0.040	19.923	15.495	0.010	16.111	14.265	0.025
At most 1	0.010	3.812	3.841	0.051	3.812	3.841	0.051

Trace test indicates 1 cointegrating eqn at the 5% level

Max-eigenvalue test indicates 1 cointegrating eqn at the 5% level